Adapting to Brexit: the Response of Corporate Structures to Geopolitical Uncertainty *

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Abstract

Geopolitical uncertainty alters firms' incentives to organise their corporate structures across borders, creating a distinct margin of adjustment to policy risk. Using firm-level data on UK–EU parent–subsidiary links from 2011 to 2021, we examine firms' responses to the prolonged uncertainty following the Brexit referendum. After the vote, UK firms established more first subsidiaries in the EU than in the equivalent period before it, while EU firms established fewer in the UK. UK firms establishing their first EU subsidiary after the vote were weaker on entry than those that had done so before; EU firms establishing UK subsidiaries were stronger. Increases in Brexit-related uncertainty were associated with more subsidiaries in both directions, particularly among small firms, with no comparable response in placebo destinations: a one-percentage-point increase in the uncertainty index was followed by about 23 additional UK-to-EU first subsidiaries over six months, compared with 12 in the opposite direction. To rationalise these findings, we propose a two-period model of organisational choice under uncertain market-access costs. Changes in expected relative costs explain the opposite patterns of subsidiary formation and selection. Because a firm that waits remains exposed to uncertain cross border costs, uncertainty itself can encourage precautionary subsidiary formation, despite sunk organisational costs when the insurance value of local presence outweighs the flexibility of serving the market from abroad and the option to switch later.

KEYWORDS: Brexit, foreign subsidiary, geopolitical uncertainty, parent firm, Brexit Uncertainty Index, precautionary FDI.

JEL CLASSIFICATION: F21, F23, G32, F15, D22

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1. Introduction

The June 2016 vote of the United Kingdom to leave the European Union marked the onset of a prolonged period of uncertainty about future market access between the UK and the EU. Firms on both sides of the Channel faced the possibility that the fixed and variable costs of serving a key foreign market might rise through new trade barriers, regulatory divergence, or changes in tax and legal regimes. One margin of adjustment was the decision to establish a corporate presence across the border, incurring fixed organisational costs to preserve market access and reduce exposure to future frictions. This paper investigates how anticipated changes in market access relate to the formation of foreign subsidiaries and the characteristics of firms establishing them.

Until the end of the transition period in December 2020, the main rules governing UK–EU market access remained in place, so firms faced changes in expected future costs and uncertainty about them rather than the full realisation of new trading frictions. Uncertainty surrounding the UK–EU economic relationship fluctuated substantially as negotiations evolved and persisted beyond the conclusion of the Trade and Cooperation Agreement (TCA) in December 2020. Survey evidence shows that Brexit remained a major source of uncertainty for a significant share of UK firms even after formal withdrawal (Bloom, Bunn, Mizen, Smietanka and Thwaites, 2025). This prolonged period of adjustment, during which firms could respond before the new trading arrangements took effect, provides a natural setting in which to study changes in their international organisational footprint. Although our empirical setting is Brexit, organisational restructuring in anticipation of changes in market access is relevant to a broader set of geopolitical shocks, including trade wars, sanctions, and episodes of regulatory fragmentation (Handley and Limão, 2017).

We use the referendum and the subsequent negotiation period to examine whether firms established subsidiaries across borders in anticipation of higher trading frictions and regulatory barriers. We focus on the creation of new parent–subsidiary links as a measure of foreign direct investment (FDI) at the extensive margin, drawing on *Orbis Historical* data covering 13,763 British firms and 10,552 EU firms between 2011 and 2021. We examine both the overall extensive margin of subsidiary formation and the formation of first subsidiaries, defined as new subsidiaries established by firms with no other subsidiary links in the preceding year.

We document three central findings. First, UK firms established 20% more first subsidiaries in the EU after the referendum than in the equivalent period before it, whereas EU firms established 4.4% fewer first subsidiaries in the UK. Second, UK firms establishing their first EU subsidiary after the referendum had weaker pre-incorporation performance, assessed through several balance sheet figures, than UK firms that had done so before the referendum, whereas EU firms establishing their first UK subsidiary were stronger than their pre-referendum counterparts. Third, fluctuations in Brexit-related uncertainty, measured by the Brexit Uncertainty Index (BUI), were positively associated with first subsidiary formation in both directions, particularly among small firms. A one-percentage-point increase in the BUI was followed by approximately 23 additional UK-to-EU first subsidiaries over six months, compared with 12 in the opposite direction. The UK response was larger and more persistent. Multiplying the additional subsidiary counts by median reported subsidiary assets gives illustrative amounts of approximately £10.7 million

for UK-to-EU subsidiaries and £2.6 million for EU-to-UK subsidiaries per percentage-point increase in the uncertainty index; these are not direct estimates of FDI flows. We also find suggestive evidence that domestic organisational expansion weakened in the UK: increases in Brexit-related uncertainty are associated with fewer new UK-UK parent-subsidiary links, although these estimates are less precise. This pattern is consistent with evidence that Brexit-related uncertainty depressed domestic investment (Bloom et al., 2025).

The empirical evidence leave us with a puzzle. The level and selection changes run in opposite directions for UK and EU firms, yet increases in Brexit-related uncertainty were associated with more subsidiary formation in both. To reconcile these patterns we complement our empirical analysis with a two-period model a la Helpman, Melitz and Yeaple (2004) in which heterogeneous firms decide whether to establish a foreign subsidiary before future market access costs are known. The model distinguishes changes in expected costs from uncertainty about those costs and shows how both affect subsidiary formation and the selection of new parent firms. The model suggests that, for UK firms, an increase in the expected cost of serving the EU market from abroad makes establishing an EU subsidiary relatively more attractive, lowering the subsidiary formation cutoff and inducing additional, weaker firms to establish a local presence. On the other hand, for EU firms the observed decline in UK subsidiary formation and stronger selection can be explained by a sufficiently large increase in the expected fixed cost of operating a UK subsidiary, relative to the cost of serving the market from abroad. Beyond these changes in expected costs, the model identifies a distinct precautionary motive for FDI: uncertainty can encourage subsidiary formation despite sunk organisational costs, because firms that postpone establishing a subsidiary remain exposed to uncertain cross border costs. This mechanism can generate positive responses to uncertainty in both directions, with a stronger response among UK firms, reflecting their greater exposure to the EU market.

Our analysis relates to several strands of the literature on the economic effects of Brexit, firms' responses to policy uncertainty, and the determinants of foreign direct investment.

First, a substantial literature examines the effects of Brexit on trade, investment, employment and aggregate outcomes. Early analyses considered the potential consequences of leaving the EU (HMTreasury, 2016; IMF, 2016; Van Reenen, 2016; Dhingra et al., 2017; Sampson, 2017), while subsequent work has examined realised and anticipated effects on trade and productivity (Born et al., 2019; Dhingra and Sampson, 2022; Haskel and Martin, 2023; Broadbent et al., 2024). Studies of exporting and trade-policy uncertainty include Crowley et al. (2018), Fernandes and Winters (2021) and Graziano et al. (2020). Breinlich et al. (2020), using a synthetic-control approach, estimate that the referendum increased UK outward investment transactions into the EU by 17% between 2016 and 2019. We examine a different margin, first subsidiary formation rather than investment transactions, and cover both UK-to-EU and EU-to-UK links, the pre-incorporation characteristics of new parent firms, within-period associations with measured Brexit uncertainty, and placebo destinations. McGrattan and Waddle (2020) study Brexit and multinational production, while Freeman et al. (2025) examine UK firms' trade after Brexit. Other studies investigate non-trade-policy barriers (Crowley and Hoste, 2025), services trade (Breinlich and Magli, 2025), local labour markets (Javorcik et al., 2025), cross-border research collaboration (Vanino et al., 2025) and monetary-policy responses (Geiger and Güntner, 2024).

Second, our findings relate to research on the effects of policy uncertainty on firm behaviour. Bloom et al. (2025) use the Decision-Maker Panel to document the persistence of Brexit uncertainty and its adverse effects on UK investment and economic activity. Hassan et al. (2024) examine firm-level exposure to Brexit uncertainty and its consequences for valuations, hiring and investment. Kumar et al. (2023) show that uncertainty affects firms' investment and operating decisions, while Bahaj et al. (2024) document unusual patterns of business creation during the COVID-19 pandemic. Our results show that the formation of first foreign subsidiaries is another margin associated with Brexit-related uncertainty.

Third, the paper contributes to the literature on heterogeneous firms, exporting and multinational activity. Melitz (2003) emphasises the selection of more productive firms into exporting, while Helpman, Melitz and Yeaple (2004) characterise the sorting of firms between exporting and foreign production when fixed costs differ across modes. Related work examines barriers to foreign-market activity (Caggese and Cuñat, 2013; Djankov et al., 2010), multinational dynamics (Gumpert et al., 2020) productivity gains from foreign investment (Fons-Rosen et al., 2021) and the macroeconomic consequences of cross-border ownership for the comovement of national markets (Anagnostopoulos, Atesagaoglu, Faraglia and Giannitsarou, 2022). Our finding of weaker pre-incorporation performance among UK parents establishing first EU subsidiaries, but stronger performance among EU parents establishing first UK subsidiaries, calls for a mechanism that allows the selection cutoff to move in opposite directions. In our model, changes in expected relative costs move the subsidiary-formation cutoff and therefore affect both subsidiary counts and selection.

Our theoretical contribution builds on the heterogeneous-firm framework of Helpman et al. (2004) and relates to the literature on multinational production and the proximity–concentration trade-off, including Ramondo et al. (2013) and McGrattan and Waddle (2020), as well as the option value of waiting under trade-policy uncertainty studied by Handley and Limão (2015). We bring these mechanisms into a common organisational decision, deriving an exact decomposition of the effects of expected costs, profit convexity, precautionary insurance and future switching. Uncertainty about proportional trade costs can favour serving a market from abroad through the convexity of operating profits, whereas a subsidiary can insure a risk-averse firm against uncertain cross-border costs. Uncertain fixed cross-border compliance costs strengthen the insurance motive because they create profit risk without the same convexity benefit as proportional trade costs. We derive analytical conditions under which the insurance value of local presence dominates the convexity benefit of cross-border provision when firms do not subsequently switch organisational form. We then use numerical simulations to examine how the option to switch later affects precautionary subsidiary formation.

The remainder of the paper is organised as follows. Section 2 describes the data and variable construction, including stylised facts on subsidiary formation. Section 3 analyses the characteristics of firms establishing their first foreign subsidiary. Section 4 studies the dynamic relationship between subsidiary formation and Brexit-related uncertainty. Section 5 presents robustness checks and additional results. Section 6 develops the model of foreign subsidiary formation under expected and uncertain costs, and Section 7 concludes.

2. Data Description and Some Stylized Facts

2.1 Orbis data. The core data source is the *Orbis Historical* database, primarily known for its extensive collection of company financial data for millions of firms worldwide, spanning several decades.[1] It is the main source of data that we use to answer our research questions of how British and European firms' corporate structures responded to Brexit as it offers comprehensive data on the ownership structure of entities, and links this with detailed firm characteristics and balance sheet data. Importantly, it provides a representative sample of the European and British economies by including small, unlisted firms (Kalemli-Ozcan, Sorensen, Villegas-Sanchez, Volosovych and Yesiltas, 2024). The good coverage of firm size will allow us to study also any heterogeneous behaviour of firms before and after the referendum, underlining the importance of a holistic analysis of the economy instead of focusing on listed firms only.

To construct our datasets, we use *Orbis Historical* data covering several parent-subsidiary configurations. Specifically, we identify British firms with subsidiaries in EU member states, EU firms with subsidiaries in the UK, and EU firms with subsidiaries in EU countries other than the ones that they are incorporated in. We additionally extract data on British firms with subsidiaries in two non-EU countries, Norway and Switzerland, to conduct placebo analyses. Finally, we collect information on British firms with domestic subsidiaries, in order to examine the effect of Brexit-related uncertainty on home-country incorporations. The data extracted are from calendar years 1998 to and including 2021.[2] Our dataset contains information on the ownership structure of entities, detailing shareholdings, subsidiaries, direct and indirect ownership shares, ultimate owners, and corporate groups.

For our study, we will use the following terms throughout the paper, defined as follows: A *parent* firm is an Orbis shareholder with direct ownership exceeding 0% in a *subsidiary* firm.[3] For both parent and subsidiary firms, we have a wide range of firm characteristics including sector, geographic location, date of incorporation, size (Orbis variable), and financial metrics, including turnover, profits, assets, and cash flow.[4]

The main unit of analysis is a *parent-subsidiary link*, which we define as the relationship of a parent firm being the immediate (direct) shareholder of a subsidiary, in a given year of Orbis data. Thus, we keep only observations where parent and subsidiary names, subsidiary date of incorporation and direct ownership share are not missing, and we also drop observations reporting a subsidiary's indirect owners, including global ultimate owners. Moreover, to ensure data consistency, we drop parent-subsidiary links where the information date of when the link is formed is missing, or where the year of link predates the year of incorporation of the subsidiary.

We are interested in the establishment of subsidiaries by parent firms, which can be proxied in the data by carefully defining the *formation of a new parent-subsidiary link*. Orbis reports the information

[1] We use the Historical Orbis Database for Europe, which comes from Bureau van Dijk (BvD), a Moody's Analytics company.

[2] Ownership data in *Orbis Historical* starts in 2007, however it is not well populated until 2008.

[3] Parent firms do not always coincide with *global ultimate owners,* who are defined as firms with indirect ownership of more than 50% of the subsidiary.

[4] Appendix B describes the cleaning procedure for the balance sheet data of firms, following Kalemli-Ozcan, Sorensen, Villegas-Sanchez, Volosovych and Yesiltas (2024).

date at which the link was reported to them, however, this does not necessarily coincide with the exact date the link was actually formed due to reporting conventions. Instead, the calendar year in which a link first appears in the dataset is a more reliable indicator. We note that the year of a link can be different from the date of incorporation of the subsidiary, for instance, when a parent company just acquires a stake in an existing company.

Since our main research question asks whether and how parent firms establish new, i.e. not pre-existing, subsidiaries in response to Brexit uncertainty, we define a link as *new* if it satisfies the following three criteria: First, the link is labelled as new only for the year that it first appears in the dataset. As an example, consider a parent firm P and subsidiary firm S, and suppose that parent-subsidiary links between P and S exist for years 2013, 2014, 2015 and 2016. Then we count the link in 2013 as a *new* link, since the link was formed in 2013, and interpret this as firm P establishing a new subsidiary in that year. The links in the following years 2014, 2015, and 2016 are recorded in the data as pre-existing, simply reflecting the fact that the parent firm P has established a subsidiary which it keeps owning up and until 2016. Second, we require that the new link is formed within one year of the subsidiary's date of incorporation (DOI) and henceforth use this DOI as our main time variable. This ensures that we truly capture the creation of new subsidiaries rather than parents acquiring stakes in an existing firm. Third, to cleanly identify the formation of a new link, we also condition on the direct ownership share of the parent firm in the subsidiary to exceed 20%.[5] We refer to a parent firm as a *new parent* and to a subsidiary firm as a *new subsidiary* when the two have formed a new link, as defined above.

We are primarily interested in understanding whether firms with no foreign subsidiaries create new links for the first time after the Brexit referendum. We call such new links between a parent and subsidiary *first new link*, and we formally define it as a new link for which the new parent does not have any other link to subsidiaries (new or pre-existing) in the year prior to the link formation. We refer to parent and subsidiary firms of such first new links as *first-time new parents* and *first new subsidiaries* respectively. By definition, the set of first new links is a subset of the set of new links in our dataset.

We apply two final filters to our baseline datasets. First, we exclude from the samples of parent firms companies that are in the finance and insurance sector. The baseline variable we use for this exclusion is NACE letter code K (financial and insurance activities)[6]. Whenever the NACE letter is not available, we use as much as possible the Orbis variables for trade description and the names of the companies to refine the sector classifications and identify more firms that are in the finance sector. Second, we exclude parent firms for which we have no information about their business activities (either because of lack of sectoral classification, or because we cannot deduce it from text and description variables) and are classified as *holdings* companies.

Applying these data filters and definitions, and restricting the sample to the period from January 2011 to December 2021, covering five years before and after the Brexit referendum in late June 2016, we construct three primary datasets of newly formed parent-subsidiary links over time, each with rich firm-level characteristics for both parents and subsidiaries. The first dataset, UK-EU, contains the firm data for all new links of non-financial parent firms headquartered in the UK that establish subsidiaries in EU

[5] We can vary the percent direct ownership and the year of link-DOI relation for sensitivity.

[6] According to NACE classification revision 2.

countries. The second dataset, EU-UK, includes all new links of non-financial parent firms from EU countries that establish subsidiaries in the UK. The third dataset, EU-EU, consists of new links between non-financial parent firms from any EU country that establish subsidiaries in EU countries other than their home country, and thus are classified as foreign subsidiaries. We also construct a placebo dataset, UK-NOCH, which includes new links between UK non-financial parent firms and subsidiaries located in Norway and Switzerland. Finally, we build an auxiliary and substantially larger dataset, UK-UK, containing all new links between UK non-financial parent firms and domestic UK subsidiaries, again with rich firm-level characteristics for both parents and subsidiaries.

2.2 Uncertainty indices and other macroeconomic variables. We gauge uncertainty through the Decision Maker Panel (DMP), a firm-level survey led by a team of researchers at the Bank of England, University of Nottingham and King's College London (Bunn, Bloom, Crundwell, Khan, Menzies, Mizen, Sculthorpo, Shah, Twaites and Yotzov, 2024). The survey results are used to create two uncertainty indexes used in our analysis: (i) Brexit Uncertainty Index (BUI) or (ii) Covid Uncertainty Index (CUI). Each index measures the percentage of firms in each month that identify Brexit/Covid-19 as one of the top three sources of uncertainty in the economy. Spanning 8,000 active UK firms with over 10 employees, the survey employs a rotating, three-panel structure to randomly select firms, ensuring each panel receives one third of the questions monthly and all firms cycle through all questions within each quarter. The BUI data series begins in May 2015, with data prior to September 2016 imputed using betting odds data. The CUI data series starts in March 2020.

In our econometric exercises, we control for several macroeconomic variables at country level that come from several sources. (i) The *Office of National Statistics UK* and *Eurostat* provide the trade data we use as controls in our analysis, specifically variables for exports from UK into EU as whole, as well as from the EU to UK. (ii) the *Bank of International Settlements* provides the real effective exchange rate, (iii) *OECD* provides the real GDP for UK and each European country, (iv) *Fred Data* provides the 3-month interbank interest rate for UK.

2.3 Data summary and stylized facts. Table 1 reports the summary figures for the five datasets we use. The cleaned UK-EU dataset consists of 18,223 new parent-subsidiary links, involving 13,763 distinct parent firms and 17,833 new subsidiaries in the EU. Most of the links, 80%, are first time links, where the parent did not have any other subsidiary in the year before the link appeared. More than half (54%) of all new links were formed after the Brexit referendum. Investigating these links over time, we find a large 17.8% increase in the number of new links formed from before to after the Brexit referendum. This increase is even larger, 20%, when considering only first-time parent-subsidiary links where parents have no prior link in $t-1$.[7] Figure 1 panel (A) shows the evolution over time of the number of new parent-subsidiary links at quarterly frequency where the purple bars denote new links and the orange bars first new links. The dotted vertical line identifies the quarter of the Brexit referendum, Q2 2016. Two facts are evident from the figure: first new links are the majority of links that have been created over the sample

[7] Pre-Brexit refers to 2011q1 to 2016q2, and post-Brexit encompasses 2016Q3 to 2021Q4. We choose the same length in quarters to ease comparison. The time dimension is the subsidiary DOI.

TABLE 1: Summary of Orbis data

	New par-subs links	New parents	New subs	First new par-subs links	First-time new parents	First-time new subs
UK → EU	18,223	13,763	17,833	14,568	13,709	14,303
EU → UK	13,269	10,552	12,899	10,857	10,517	10,557
EU → EU	77,492	58,214	74,297	62,477	58,022	60,081
UK → NOCH	907	850	893	873	850	861
UK → UK	1,218,191	1,025,715	1,181,460	1,040,743	1,024,451	1,027,680

Notes: Period 2011m1-2021m12. Excludes parent firms from sectors 'financial' and 'holdings'.

period accounting over all and across time. Additionally, an upward trend can be identified in the link formation. The referendum does not seem to have broken the upward trend. The only temporary break in the trend can be attributed in the second and third quarter of 2020 during the first period of the COVID pandemic.

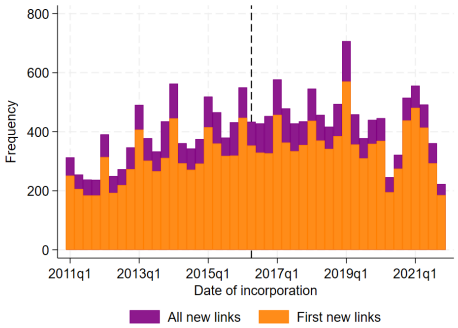
Similarly, 10,552 EU non-financial parent firms established 12,899 subsidiaries in the UK between Jan 2011 and Dec 2021, creating 13,269 new links. About 48% of these links are formed post-referendum, a smaller proportion than for UK-EU. Investigating the time series of links, we find the opposite pattern than for UK-EU: The average number of new links formed decreases by 7.8% from before to after the Brexit referendum. Interestingly, this number is slightly smaller for first-time links, decreasing only by 4.4%. This is also evident from Figure 1 panel (B), where our data suggest that the number of links followed a flat or slightly negative trend from 2016 to 2019.

We now turn to the three datasets to be used as placebo or control groups of counterfactual exercises in our empirical analysis. The first is the dataset for links between European parents and European subsidiaries, which is larger, as it encompasses multiple countries on both sides of the relation, with 77,492 new parent-subsidiary links. As such, we find 58,214 distinct non-financial European parents, with 74,297 new subsidiaries are not incorporated in their own country. 53% of these links were formed after the Brexit referendum, an increase of 14.2% from the number of links formed before the referendum. Interestingly, if we zoom in only on first links, the growth rate is lower, 9.4%, and half of the increase in first links created by UK parents in the EU. The time series for new incorporations of European subsidiaries by European parents is relatively stable before and after Brexit, showing a positive trend over our sample period (see Figure A.1 panel (A), in the Appendix).[8] The second dataset includes links between the UK and Norway and Switzerland. There are only 907 links from non-financial British parents to Norwegian and Swiss subsidiaries in our full sample. The number of links decreases substantially from before to after

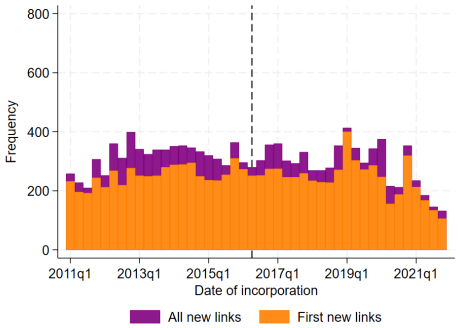
[8]For some of the analysis, we use a subset of the EU-EU sample which excludes European firms that also have subsidiaries in the UK, i.e. they appear in our EU-UK sample. There are 1,077 such European firms, i.e. less than 2% of the 58,214 distinct non-financial European parent firms.

FIGURE 1: New links and first new links

(A) UK → EU



(B) EU → UK



Notes: New parent-subsidiary links created by a UK parent in an EU country (panel (A)) and in UK by EU parents (panel (B)). Purple bars denote all new links and orange bars all first new links. The dotted vertical line marks 2016q2, identifying the quarter of the Brexit referendum. Time period 2011q1-2021q4 and frequency is quarterly.

Brexit, -48.3%. This trend holds both for Norway and Switzerland, such that including them in the same sample does not mask any idiosyncratic trends (Figure A.1 panel (B) in the Appendix). The third dataset includes all domestic parent-subsidiary links, formed between UK parent firms and UK subsidiaries. It is the largest of all the datasets considered and contains 1,218,191 new subsidiary links, involving 1,025,715 new parents and 1,181,460 new subsidiaries. Of the new links, 1,040,743 are first new links (about 85% of all new links), with 1,024,451 first time new parents, and 1,027,680 first new subsidiaries. We find a decline in link formation following the referendum for this sample (Figure A.1 panel (C) in the Appendix). On average, the number of new links and first new links decreased by 14.1% and 8.5% respectively relative to the pre-referendum period.

We also explore the number of new links by splitting the sample by the size of the parent firm. In Figure 2, the left column shows the UK-EU links by parent size and the right column EU-UK links. The top row shows all new links and the bottom row first new links. First we note that small parents have mostly created new links for the UK-EU, 63%, however this proportion is considerably smaller for the EU-UK dataset, 47%. The UK-EU links of parents of any size have increased after the referendum, apart from the ones created by very large parents. In contrast, we see a decrease in links for parents of any size in the EU-UK sample. If we zoom into the first new links, the picture is similar in both samples. Small UK parents create predominately EU links and medium firms are the ones that show the larger increase after Brexit. Interestingly, the first new UK links of small EU parents are the only ones that slightly increase after the referendum.

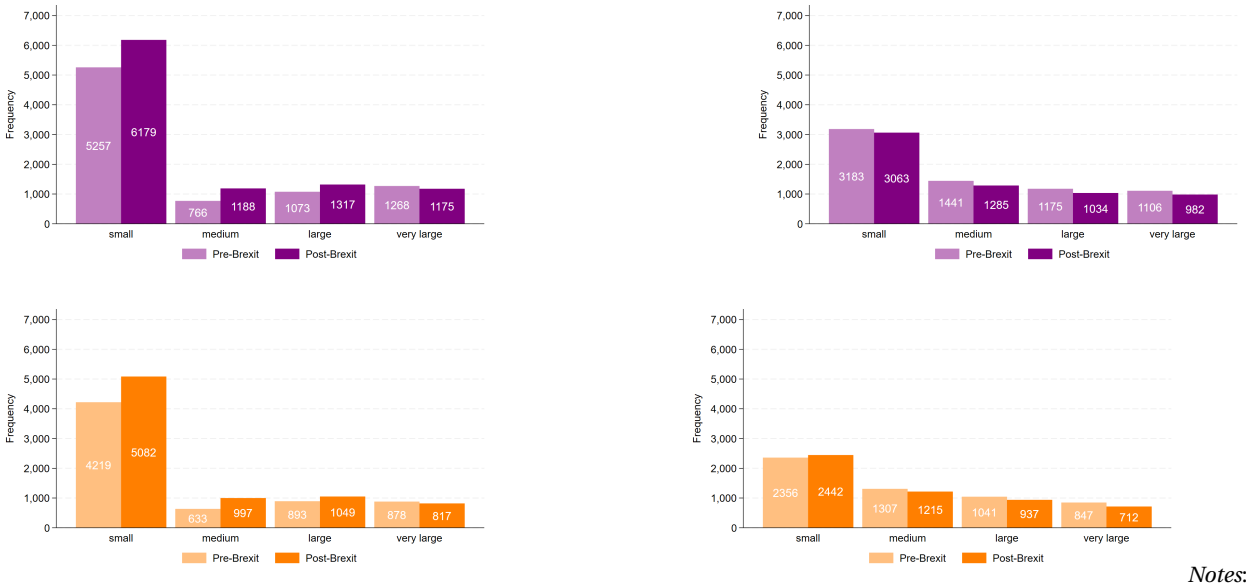
3. Which Types of Firms Establish New Subsidiaries Post-Brexit Referendum?

Our findings so far suggest an increase in subsidiaries established by UK firms in the post referendum period compared to the pre referendum one; in particular, small parents have increased significantly their foreign incorporations, although all size groups have shown an increase. In contrast, we do not see the same pattern in EU-UK incorporations but the opposite both at the aggregate level and when we

FIGURE 2: New links and first new links by parent size

(A) UK → EU

(B) EU → UK



Notes: New parent-subsidiary links created by a UK parent in an EU country (column (A)) and in UK by EU parents (column (B)). Top row shows all new links and the bottom row first new links. Parent size follows Orbis definition. Time period 2011Q1-2021Q4, pre referendum period 2011Q1-2016Q2 and post referendum 2016Q3-2021Q4.

focus on size groups. In this section, we analyse various characteristics of parents around the time that they decide to establish a subsidiary.

For the rest of the paper we refine our sample in two dimensions. Firstly, we concentrate our analysis to first new links. As discussed in the previous section, first new links are around 80% of all new links in all our samples. This sample restriction will help us study the decision of firms to become multinationals. Second, we exclude from the EU-EU sample any EU firms that, apart from subsidiaries in other EU countries, also have foreign subsidiaries in the UK during our sample period (i.e. they also belong in the EU-UK sample). This restriction allows us to construct a control group that is as much as possible insulated from the uncertainty and structural changes associated with Brexit and more generally the UK.

We start with the premise that typically companies that establish subsidiaries tend to be relatively productive and experience growth (Melitz, 2003; Helpman et al., 2004), and ask whether the uncertainty surrounding future trade policy may prompt a different type of firms to establish a first subsidiary. To investigate this, we focus on the balance sheet characteristics of parent firms in our samples *one* and *two* years prior to establishing their first foreign subsidiary in UK or EU, as indicators of their financial and corporate 'health' before their decision to do so.

We conduct a difference-in-difference analysis using repeated cross-sectional yearly data from 2011 to 2021 to assess and compare the health of such UK and EU parent firms before and after the Brexit

referendum. To this end, we estimate the following two baseline specifications:

$$Y_{-2,p} = \alpha_{-2} + \beta_{-2}(POST_p \times UKEU_p) + \sum_{j=1}^{3} \gamma_{-2}^{j} X_{-3,p}^{j} + \varepsilon_p, \tag{1}$$

$$Y_{-1,p} = \alpha_{-1} + \beta_{-1}(POST_p \times UKEU_p) + \sum_{j=1}^{3} \gamma_{-1}^{j} X_{-2,p}^{j} + \varepsilon_p, \tag{2}$$

where p indexes parent firms. We denote with $Y_{-1,p}$ and $Y_{-2,p}$ the firm dependent variables of interest, as recorded one and two years respectively before the year of incorporation of the first subsidiary of p, in the given sample. The dependent variables we consider are the growth rates of (i) turnover, (ii) sales and (iii) cash flows, and the changes in (iv) return on equity, (v) profits-to-assets ratio and (vi) leverage.[9] The variable $POST_p \times UKEU_p$ represents the treatment, and is the product of a dummy variable $POST_p$ equal to 1 if the subsidiary is incorporated after the Brexit referendum date, and a dummy variable $UKEU_p$ equal to 1 if the parent firm is in the UK-EU dataset and 0 if it is in the EU-EU dataset. In other words, the treatment variable is 0 if the British parent company established the subsidiary before the referendum, or the parent company is from an EU country and has established a first subsidiary in another EU country before the referendum.[10] We also include three aggregate controls that are lagged relative to the year of incorporation of the first subsidiary and denoted $X_{-3,p}^{j}$ and $X_{-2,p}^{j}$ respectively, in order to capture relevant economic conditions in the country of the parent company that may affect the decision of the company to expand with subsidiaries abroad. The controls we include are the growth rates of real GDP, real exports and real effective exchange rate, all for the country of the parent company.[11] Finally, ε_p is the residual term and we cluster standard errors at the parent country level.

Table 2 shows the estimated coefficients of interest β_{-2} and β_{-1} from specifications (1) and (2). Across the parent-firm characteristics we examine, our results point to an interesting pattern: UK firms establishing a first subsidiary after the Brexit referendum, when uncertainty about the future trade relationship was high, tended to exhibit weaker performance than non-treated firms. Specifically, such UK firms experienced about 8 percentage points lower turnover growth, and 4 percentage points lower sales growth in the year preceding their first subsidiary incorporation. These firms were also more leveraged and had a significant drop in their return on equity. Most of these results remain true when we consider the same characteristics two years before the incorporation of the subsidiary, when we see that firms have a de-

[9] The firm balance sheet data is from *Orbis Historical*, all expressed in Euros. For the analysis, we limit our sample to parent firms that report unconsolidated balance sheet data, which constitutes 89% of the sample of UK firms and well over 80% of the sample of EU firms. Prior to doing the regressions, the data was cleaned and prepared following the principles of Kalemli-Ozcan et al. (2024) of how to handle balance sheet data from *Orbis Historical*.

[10] In the Figure A.2 in the Appendix, we provide the distribution of the balance sheet variables of interest for the treated and control groups. We do this to first assess if the distribution of dependent variables follows a normal shape, and second to provide an initial understanding of the sources of differences among the groups. Figure A.3 provides the same information for the dependent variables of the EU-UK analysis. The large spikes at the tails of the distributions are due to the fact that variables are winsorised at the 5 and 95 percentiles, as is common when using firm data.

[11] We note that the variables X are not firm specific, but macroeconomic aggregates. The notation p is used only to map the variable to the country of the parent firm. The sources for all macro aggregate variables we use are *OECD data*, *Bank of International Settlements*, *Eurostat*, and *Office of National Statistics UK*.

TABLE 2: Balance sheet characteristics of UK parent firms with first subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-4.146**	-8.158***	-7.051***	-3.841**
	(1.606)	(1.658)	(1.844)	(1.737)
Observations	8,718	9,629	6,628	7,251
R^2	0.002	0.003	0.002	0.001
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-0.449	-2.248***	1.887	-3.869
	(0.319)	(0.405)	(2.852)	(3.703)
Observations	9,398	10,287	8,609	9,371
R^2	0.004	0.002	0.001	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.443**	-0.230	0.905***	0.274
	(0.175)	(0.182)	(0.169)	(0.215)
Observations	10,462	11,501	9,015	9,744
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical*, *Office of National Statistics UK*, *Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

creased profits-to-assets ratio as well. These findings align with the research by Kumar et al. (2023), who show that firms experiencing increased uncertainty tend to reduce employment and witness a decline in sales.

The baseline sample is defined so that the non-treated group consists of EU parent firms that had already incorporated an EU subsidiary before the Brexit referendum only. This choice of control group addresses a potential concern that, following the referendum outcome, some EU firms may have been affected by the shock and responded by expanding into other EU countries rather than the UK. Nevertheless, when we set aside this concern about potential self-selection and include EU parent firms that established an EU link after the referendum as an additional control group, the results remain robust and become even stronger across all dimensions (see Table A.2 in the Appendix, for this full sample of the control group). This suggests that post-Brexit UK parent firms perform even more weakly when compared with this expanded control group.

Next, we repeat the same analysis considering now EU firms that incorporate a subsidiary in the UK after the referendum, using the similarly defined control group as before, i.e. EU parents that have established a first subsidiary in the UK and EU parents that established a first subsidiary in another EU country

TABLE 3: Balance sheet characteristics of EU parent firms with first subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.606***	8.197***	12.002***	6.705***
	(2.536)	(2.247)	(2.146)	(1.313)
Observations	9,073	10,098	8,306	9,143
R^2	0.004	0.002	0.006	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.925***	-1.160	11.409**	11.674
	(1.350)	(1.357)	(4.928)	(7.267)
Observations	9,916	10,959	9,070	9,899
R^2	0.005	0.002	0.001	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.450***	-0.209	-0.225	-0.125
	(0.509)	(0.616)	(0.187)	(0.326)
Observations	10,950	12,139	9,209	10,003
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

before the referendum, estimating the following two specifications:

$$Y_{-2,p} = \alpha_{-2} + \beta_{-2}(POST_p \times EUUK_p) + \sum_{j=1}^{3} \gamma_{-2}^{j} X_{-3,p}^{j} + \varepsilon_p, \tag{3}$$

$$Y_{-1,p} = \alpha_{-1} + \beta_{-1}(POST_p \times EUUK_p) + \sum_{j=1}^{3} \gamma_{-1}^{j} X_{-2,p}^{j} + \varepsilon_p, \tag{4}$$

where the notation is the same as in equations (1)-(2). Here, the treatment is the product of $POST_p$, and a dummy variable $EUUK_p$ equal to 1 if the parent firm p is based in an EU country and its first subsidiary is incorporated in the UK. The results reported in Table 3 present a markedly different picture from the previous sample. Most estimates exhibit the opposite sign relative to the earlier exercise and are statistically significant. EU parent firms that incorporated their first subsidiary in the UK after the referendum appear to be healthier than EU parent firms that either undertook the same expansion before the referendum or incorporated a subsidiary in another EU country prior to the vote. When we further expand the control group to include EU parent firms that established subsidiaries after Brexit, the results remain qualitatively unchanged, although somewhat attenuated (see Table A.3 in the Appendix).

Taken together, these findings suggest a clear asymmetry in post-referendum incorporation behaviour. UK parent firms establishing subsidiaries in the EU may have done so primarily as a defensive response to the anticipated frictions associated with Brexit, rather than as part of an expansion strategy grounded in strong expected performance. This is consistent with their weaker pre-incorporation out-

comes relative to comparable EU firms. By contrast, EU parent firms that chose to establish a subsidiary in the UK after the referendum exhibit stronger observable characteristics than relevant control groups, indicating positive selection. Although the overall rate of EU incorporations in the UK declined modestly after the referendum, those firms that did enter the UK market do not appear to be driven by precautionary motives and instead resemble, or even outperform, firms that made similar location choices prior to the referendum.

4. Dynamic Response of First New Links to Brexit Uncertainty

The Brexit referendum introduced an extended period of uncertainty regarding the future trading relationship between the United Kingdom and the European Union. During this time, firms had to weigh the costs and benefits of establishing new subsidiaries across borders while awaiting clarity on post-Brexit trade arrangements. In this section, we examine the dynamic relationship between Brexit-related uncertainty and the monthly number of new subsidiaries established by UK firms in the EU and by EU firms in the UK.

Figure 3 plots the evolution of the Brexit Uncertainty Index (BUI) (right y-axis), i.e. the share of surveyed UK firms reporting Brexit as one of their top three sources of uncertainty, alongside the monthly number of first new links (left y-axis). Panel (A) shows first new links formed by UK firms in EU member states, while Panel (B) shows the first new links by EU firms in the UK. The 2016 Brexit vote triggered a substantial, wide-ranging, and persistent increase in economic uncertainty among UK firms. This uncertainty intensified around major political and negotiation milestones in 2018 and early 2019, and remained elevated throughout 2019 as the UK's departure from the EU was repeatedly delayed. A sustained decline only began after the December 2019 general election and the UK's formal exit from the EU in January 2020. However, Brexit-related uncertainty did not fall consistently below earlier levels until the second quarter of 2021, when the new trade agreement came into force.[12] A visual inspection suggests a positive co-movement between perceived uncertainty and link formation for both UK-EU and EU-UK samples. Periods of increased uncertainty, such as the second half of 2018, coincide with noticeable increases in the number of subsidiary incorporations. The correlation appears weaker in 2020 and 2021, likely reflecting the overlapping effects of the COVID-19 pandemic and the implementation of the final Brexit trade arrangements. Nevertheless, the spike in uncertainty at the end of 2020 is followed by a delayed rise in new EU links by UK firms in early 2021.

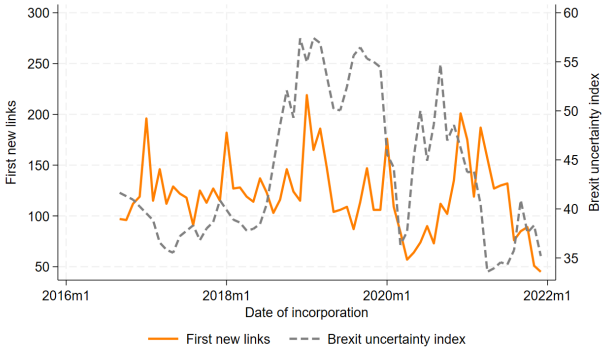
To quantify this relationship, we estimate regressions of the monthly number of first new links established by parent firms on lagged uncertainty and macroeconomic conditions. We concentrate on the number of first new links, because it captures more cleanly the extensive margin decision of firms to become multinationals. The time series covers the time period from September 2016 to December 2021, when the BUI is available.[13] For each of the two samples UK-EU and EU-UK, the baseline regression we

[12] From 2021 onward, uncertainty gradually eased but remained meaningful, affecting roughly 16% of firms, even into 2025, a pattern mirrored by trends in media coverage.

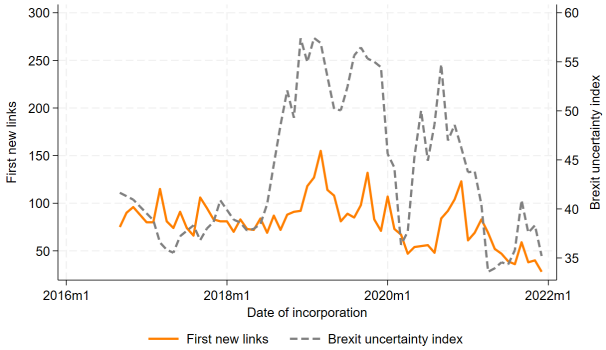
[13] As in Bloom et al. (2025), we use a single measure of uncertainty that will capture simultaneously the first and the second moment effect of Brexit.

FIGURE 3: First new links and Brexit Uncertainty Index

(A) UK → EU



(B) EU → UK



Notes: First new links over time, for UK-EU sample in panel (A), and EU-UK sample in panel (B), both plotted on the left y-axis, against the Brexit Uncertainty Index, on the right y-axis. Time period 2016m9-2021m12, data sources *Orbis Histotical* and *Decision Maker Panel, Bank of England*.

estimate is

$$Y_t = \alpha + \beta_1 BUI_{t-1} + \beta_2 CUI_{t-1} + \sum_{j=1}^{3} \gamma_j X_{t-1} + \delta_1 \mathbf{1}_t(m1) + \delta_2 \mathbf{1}_t(m8) + \sum_{j=2016}^{2021} \zeta_j \mathbf{1}_t(j) + \varepsilon_t, \tag{5}$$

where the dependent variable Y_t is the number of first new links created in month t. BUI_{t-1} is the Brexit Uncertainty Index lagged by one month, and CUI_{t-1} is a similar index, the Covid Uncertainty Index, capturing pandemic-related uncertainty. We include the same three aggregate controls as in the firm level regressions, namely the growth rates of real GDP, real exports and real effective exchange rate, all for the country of the parent firms in the sample. Dummy variables for months January and August, as well as year dummies are included to control for seasonality in subsidiary incorporation and broader temporal trends.

Table 4 reports the main estimates. Columns (1) and (2) show results for UK-EU and EU-UK subsidiary formation, respectively. The coefficient on the Brexit Uncertainty Index is positive and statistically significant in both cases. When we consider UK-EU first new links, a one-percentage-point increase in the share of firms reporting Brexit as a major uncertainty source is associated with approximately three new EU subsidiaries in the following month. The magnitude of the effect is slightly smaller for new incorporations in the UK by EU non multinational parents. While the Brexit Uncertainty Index is constructed using responses from UK firms, it can be interpreted as an upper bound on Brexit-related uncertainty for EU parents. These firms were exposed to uncertainty surrounding the future UK-EU relationship, but continued to benefit from full access to the EU's internal market, implying that their effective uncertainty level may have been lower.

Columns (3) and (4) of Table 4 show the estimated effects for the two placebo samples, EU-EU and UK-NOCH. Column (3) shows no significant relationship between Brexit uncertainty and new EU-EU new incorporations, suggesting that the uncertainty shock was specific to the UK-EU interface. Column (4) confirms that UK first incorporations in Norway and Switzerland (a placebo group) are unrelated to the BUI, underscoring that the observed effect is not a general feature of global or European uncertainty.

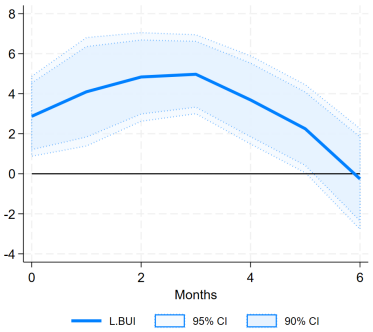
TABLE 4: First New Links and Brexit Uncertainty Index

	(1) UK → EU	(2) EU → UK	(3) EU → EU	(4) UK → NOCH
BUI_{t-1}	2.872***	2.245***	3.061	-0.016
	(0.994)	(0.406)	(2.290)	(0.091)
CUI_{t-1}	0.371*	-0.173	0.537	0.035**
	(0.213)	(0.109)	(0.836)	(0.014)
gr_{t-1}^{GDP}	0.262	-0.339	-3.359	0.090
	(0.891)	(0.868)	(5.685)	(0.063)
gr_{t-1}^{REER}	-0.947	1.948*	4.907	-0.026
	(0.979)	(0.998)	(5.718)	(0.101)
gr_{t-1}^{EX}	0.922*	0.587**	4.853**	0.000
	(0.542)	(0.289)	(2.297)	(0.000)
$Constant$	-41.656	-14.620	297.966***	5.559
	(50.401)	(17.893)	(98.175)	(4.315)
Year dummies	✓	✓	✓	✓
Jan/Aug dummies	✓	✓	✓	✓
Observations	64	64	64	64
R^2	0.600	0.701	0.893	0.268

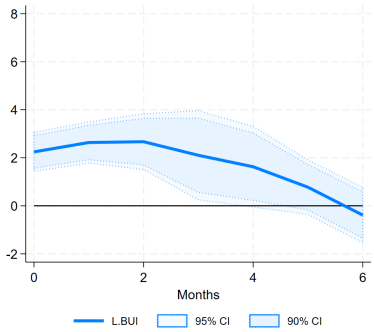
Notes: Regressions on number of creation of first new subsidiaries (first new links) on lagged Brexit uncertainty BUI_{t-1}, for four samples UK-EU, EU-UK, EU-EU and UK-NOCH, specification (5). All regressions include Jan, Aug month seasonality dummies and year dummies. Monthly frequency, period 2016m9-2021m12. Row 2 shows estimates for CUI_{t-1}. Row 3 shows estimates for lagged real GDP growth rates of the parent firm country. Row 4 shows estimates for real effective exchange rate for the countries of the parent firms. Row 5 shows estimates for lagged growth rates of exports of parent firm country. Time period 2016m9-2021m12, data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Robust standard errors. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

FIGURE 4: Dynamic responses of first new links to uncertainty

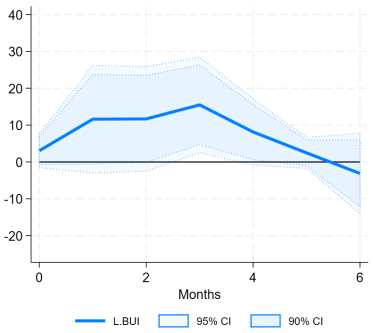
(A) UK → EU



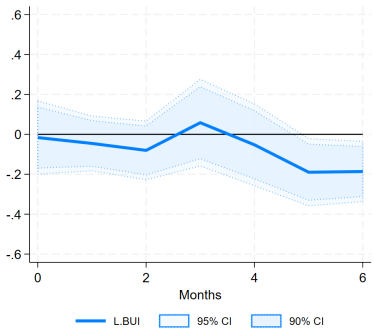
(B) EU → UK



(C) EU → EU



(D) UK → NOCH



Note: Local projection estimates of first new links by parent size on (i) BUI_{t-1}, (ii) CUI_{t-1}, (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummies, (vi) Jan and Aug month dummies (specification 6). Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*.

Covid-related uncertainty, proxied by the CUI, does not appear to influence incorporation behaviour in most of the samples, whereas export growth in the home parent country plays a positive role.

The regressions identify a correlation between one month lagged uncertainty and subsidiary formation. However, the decision by a firm to establish a foreign subsidiary and carry out the implementation of this expansion may take longer. Firms typically require time to gather information, identify a location, assess costs, and complete registration. We therefore estimate the following local projections (Jorda, 2005) to trace the dynamic response of subsidiary creation to uncertainty:

$$Y_t^h = \alpha^h + \beta_1^h BUI_{t-1} + \beta_2^h CUI_{t-1} + \sum_{j=1}^{3} \gamma_j^h X_{t-1} + \delta_1^h \mathbf{1}_t(m1) + \delta_2^h \mathbf{1}_t(m8) + \sum_{j=2016}^{2021} \zeta_j^h \mathbf{1}_t(j) + \varepsilon_t, \tag{6}$$

for $h = 0, 1, ..., 6$ months ahead. Figure 4 presents the estimated effects β_1^h. Panel (A) shows the estimates for the first new links in the UK-EU sample, where we see that the effects of Brexit uncertainty persist for several months, with a peak effect at three months after the reported Brexit uncertainty, before it dissipates after five to six months. At the peak, a one percentage point increase of BUI_{t-1} is associated

with the creation of five (4.97) first new subsidiaries by UK firms. The cumulative effect over six months corresponds to approximately 23 first subsidiaries. To put these numbers in perspective, if 1 out of 100 additional firms surveyed in a given month declared Brexit as one of their top three concerns, this is associated with a six-month cumulative increase of 23 first new subsidiaries. This translates to a 3.1% increase relative to the average six-month number of 736 first new subsidiaries in the period of estimation.[14] Our dataset for the UK-EU allows us to also provide a value calculation of the 6-month cumulative effect of one percentage point of BUI increase, to gauge the value of expansion of foreign investment due to Brexit uncertainty, as follows: the median value of yearly total assets of the subsidiaries in the UK-EU sample is about €468,300, so that the implied cumulative increase in subsidiary value due to one unit of BUI over six months is €10.7 million.

Panel (B) shows the response for EU-UK first new links. The dynamic profile is flatter, peaks at two months and fades after three months, with a cumulative effect roughly half as large, at approximately 12 first new subsidiaries for a one percentage point increase in Brexit uncertainty. A one percentage point increase in the BUI is associated with an approximate 2.4% increase relative to the average six-month number of 492 first new subsidiaries in the period of estimation.[15] In terms of value, the median value of yearly total assets of the subsidiaries of the EU-UK sample is approximately €216,800, which translates into €2.6 million per units of BUI.

For EU-EU incorporations, shown in panel (C) of Figure 4, we do not find strong evidence of a statistically significant dynamic response. However, the estimated effects are positive across horizons and reach weak statistical significance at the three-month horizon. This pattern suggests that uncertainty may have encouraged some EU parent firms to reallocate subsidiary formation within the EU rather than expand into the UK.

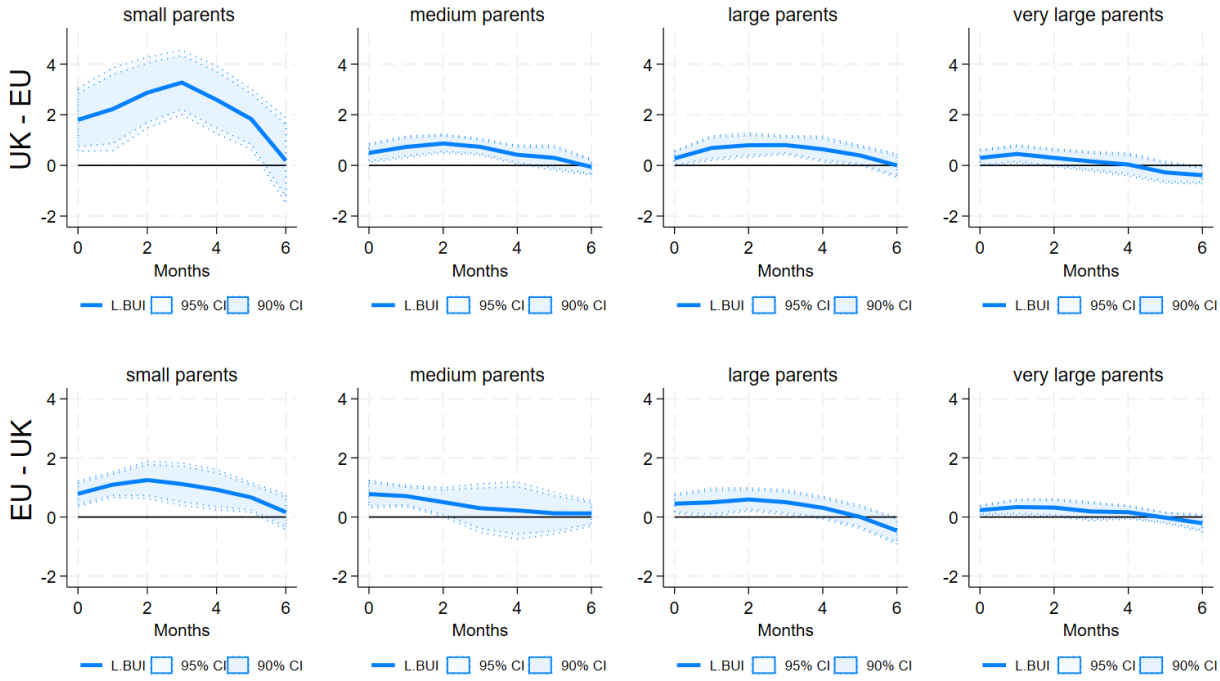
Finally, a placebo analysis using incorporations of new subsidiaries of UK parent firms in Norway or Switzerland, shown in column (4) of Table 4 and 4 panel (D) confirms that these dynamics are specific to Brexit-related uncertainty rather than a general European trend of substitution of markets for UK parents. Incorporations in non-EU European countries were unaffected by fluctuations in the BUI, consistent with the idea that the uncertainty shock was tied to the redefinition of UK-EU relations.

4.1 Results by parent firm size. Next, we explore whether the sensitivity of subsidiary formation to Brexit uncertainty varies by parent firm size. Figure 5 decomposes the impulse responses using the *Orbis* size classification. The top row shows the dynamic responses for the UK-EU sample and the bottom row shows the responses for the EU-UK sample. The results suggest that small parent firms, shown in the first column of the figure, account for the bulk of the response. Among UK parents, the number of first new links for small parent firms has the largest and most persistent reaction, with the peak occurring three months after the uncertainty shock. Medium and large firms also respond, but the cumulative effects are less than half as large, while very large firms display only short-lived and statistically insignificant responses. The results in the bottom row of Figure 5 confirm that small EU parents have also been more responsive to Brexit uncertainty than bigger firms, but the response has been flatter and more short-lived.

[14] If we consider the 90% confidence intervals, the range of percent increase is 1.3% to 4.8%.

[15] Considering again the 90% confidence intervals, this translates to an increase in the range between 0.9% and 3.8%.

FIGURE 5: Dynamic responses of number of first new links to uncertainty, by parent size.



Notes: Top row shows IRFs for UK-EU sample, and bottom row shows IRFs for EU-UK sample. LPs of first new links by parent size on (i) BUI_{t-1}, (ii) CUI_{t-1}, (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, *Office of National Statistics UK*, and *Eurostat*.

Cumulatively, the response of the number of first new subsidiaries by small parent firms accounts for approximately two thirds (65.8%) of the overall response for the UK-EU sample and approximately half (51.4%) of the overall response for the EU-UK sample. This heterogeneity highlights that smaller firms, which typically face higher fixed costs of cross-border operations, appear to have been more proactive in mitigating Brexit-related risks through the establishment of EU subsidiaries.

4.2 Establishing subsidiaries domestically. Our results so far point to an increase in the creation of new foreign subsidiaries, both from the UK into the EU and from the EU into the UK, in response to increases in Brexit-related uncertainty. This pattern suggests that firms adjusted their international organizational structures as uncertainty rose, potentially as a way to preserve market access, manage regulatory risk, or reallocate activities across borders. We now extend this analysis by complementing and contrasting these findings with evidence on the domestic formation of parent-subsidiary links within the UK.

A growing body of evidence indicates that Brexit uncertainty has been associated with a marked contraction in UK investment overall. Bloom et al. (2025) estimate that aggregate investment declined by between 12% and 16% by 2025 following the referendum, with the negative effect accumulating gradually over time and uncertainty accounting for a substantial share of the overall decline. Similarly, Keiller

(2024) documents a 7% reduction in investment in the UK manufacturing sector in the post-referendum period. Taken together, these findings suggest that, while some firms may have responded to Brexit uncertainty by expanding their foreign footprint, domestic investment activity within the UK weakened.

Against this backdrop, we turn to our final dataset, UK-UK, which records all parent-subsidiary links formed between UK parent firms and UK subsidiaries over the period January 2011 to December 2021. Our earlier analysis focused on the response of foreign investment to Brexit uncertainty. By the same token, the evolution over time in the number of (first) new links between UK parents and UK subsidiaries can be interpreted as a proxy for home investment activity. The UK-UK dataset therefore allows us to estimate directly the impact of Brexit-related uncertainty on domestic organizational expansion within the UK.

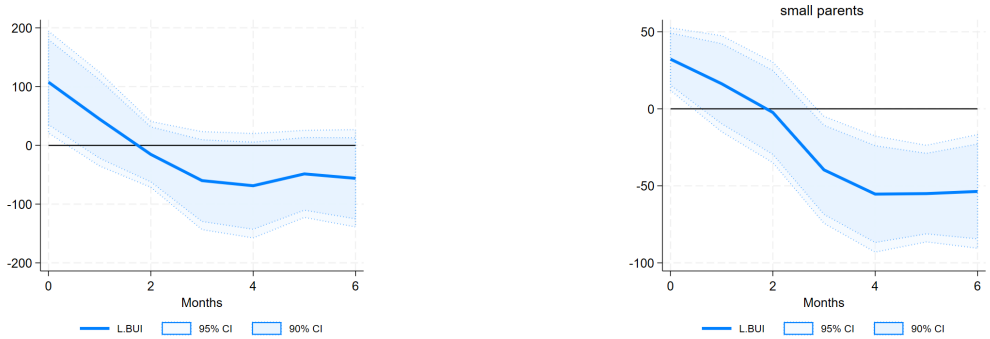
Panel (A) of Figure 6 presents the local projection estimates based on the sample of first new UK domestic links. The cumulative response is negative throughout the horizon considered, and the direction and magnitude of the estimated effect are consistent with the adverse impact of Brexit uncertainty on domestic investment documented in the existing literature. We note that the decline in the number of domestic subsidiaries primarily comes from small parent firms that refrain from expanding within the UK: Panel (B) shows the number of first new domestic links, for small UK parent firms, when we consider the subsample in which each first subsidiary reports some business activity, in the form of total assets or operating revenue, at least once between the one year before and one year after the link with the parent firm.[16]

To facilitate comparison with our estimates for foreign subsidiary formation in the UK-EU and EU-UK cases, we also compute that a one percentage point increase in the BUI is associated with a cumulative reduction of 97 new domestic subsidiaries over a six-month horizon. Given a median value of annual total assets of approximately €11,300 for UK subsidiaries in our sample, this translates into a reduction in domestic asset creation of around €1.1 million per percentage point increase in BUI. This contrast between declining domestic link formation and rising cross-border subsidiary creation underscores the reallocation dimension of firms' responses to Brexit-related uncertainty.

Overall, the evidence in this section shows that Brexit-related uncertainty did not simply depress investment activity, but reshaped it where firms chose to expand. Increases in uncertainty were associated with a persistent rise in cross-border subsidiary formation, particularly from the UK into the EU, alongside a weakening of domestic subsidiary creation within the UK. This asymmetric response reflects firms' differential exposure to the risk of losing frictionless access to the EU Single Market: UK firms faced the prospect of heightened trade and regulatory barriers, whereas EU firms retained access to their larger domestic market. The response was economically meaningful, unfolded over several months, and was concentrated among smaller firms, which typically have fewer alternative channels for maintaining market access. Rather than waiting for policy resolution, firms responded pre-emptively by reorganising their corporate structures across borders. We interpret this pattern as evidence of *precautionary FDI*, whereby firms reallocated organisational expansion internationally as a hedge against future trade and regulatory frictions.

[16] This is done in order to capture actual, productive activity by subsidiaries, rather than 'paper' or 'shell' entities. See more details on this in Section 5.

FIGURE 6: Dynamic response of domestic first new links to uncertainty



(A) UK → UK, first new links

(B) UK → UK, reporting subsidiaries

Notes: LPs of first new links by parent sector on (i) BUI_{t-1}, (ii) CUI_{t-1}, (iii) macro controls (growth rate of real GDP of UK, real effective exchange rate of UK, 3-month interbank rate) (v) year dummy variables, (vi) Jan and Aug month dummies. Panel (A) shows estimates for number of first new subsidiaries; Panel (B) shows estimates for number of first new subsidiaries of small parents, restricting to sample of subsidiaries that report balance sheet data. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, and *Office of National Statistics UK, FRED data*, and *Bank of International Settlements*.

5. Robustness and Further Results

The results are robust to a wide range of sample refinements. We first present the robustness exercises for the difference-in-differences (DiD) estimations and then those for the local projections (LP) estimations, in relation to the foreign subsidiary activity of our two main samples, UK-EU and EU-UK. In both cases, we report results for several carefully considered subsamples of the data, as we explain below. For reference, Table A.1 in the appendix reports the information on the subsamples we consider.

5.1 Balance sheet characteristics of parent firms that establish foreign subsidiaries. We repeat the DiD analysis on three refinements of the samples: (i) one that consists of parent firms that established their first and *only* subsidiary during the sample period 2011-2021, (ii) one that excludes the years 2016 and 2017 and (iii) one that excludes parent firms from an unknown sector. Below we describe each of the subsamples, and if and how the results change relative to the samples of first new subsidiaries.

First, our baseline sample consists of firms that created their first new link within the sample period but some of these firms may have created more than one link during this time. For example, a first subsidiary in 2016 followed by a second one in 2018. For these firms, the decision to expand into a multinational structure with multiple subsidiaries may have been driven by a combination of both Brexit-related considerations and other business and operational motives. In contrast, firms that became multinationals by establishing only one subsidiary during the period are more likely to have been influenced primarily by the uncertainty induced by Brexit. This restricted sample accounts for 79% for the UK-EU sample and 88% for the EU-UK sample of the first new links in the corresponding baseline samples. Table A.4 reports the results for UK parent firms creating their first and only link in the EU. These parents exhibit even weaker performance across several dimensions, such as growth in turnover, sales, as well as changes in leverage, than the parents in the baseline sample reported in Table 2. This pattern suggests

that firms opening only one subsidiary after the referendum were not scaling up an already strong business, but rather engaging in a precautionary response to heightened uncertainty. Consistent with the baseline estimations, the results for EU parent firms that incorporated their first and only subsidiary in the UK, presented in Table A.5, reveal no significant difference than the results presented in Table 3.

Second, the years 2016 and 2017 were characterised by a sharp depreciation of the British pound following the referendum outcome, which may have adversely affected the financial position of parent firms in the sample. Although the baseline specification controls for changes in UK competitiveness using the UK real effective exchange rate, we further assess robustness by restricting the sample. Tables A.6 and A.7 in the Appendix report the corresponding results and confirm the robustness of the findings. The results for both specifications are in line with our baseline estimates.

Third, we exclude from the sample any parent firms whose sector is classified as 'unknown' by our procedure, which reduces the sample to approximately 69% of the original set of first new links. Excluding these firms is potentially important because this category may include financial subsidiaries or holding companies that cannot be reliably identified during the classification and data-cleaning process. We note that a larger share of parent firms fails to report sector information in the UK than in the EU, as shown in Table A.1. The corresponding estimates are reported in Tables A.8 and A.9 for the UK and EU, respectively, and are robust to this difference in reporting. The estimated coefficients are very close to those in the baseline specification, confirming that UK firms establishing their first subsidiaries in the EU after the referendum exhibited weaker pre-establishment financial health relative to firms that did so before the referendum.

5.2 Dynamic response to Brexit uncertainty. We next turn to robustness exercises for the local projections. In particular, we assess the robustness of the results using alternative samples from two groups. First, with respect to the characteristics of the parent firms, we consider (i) the sample of *all* new links formed between 2016 and 2021, (ii) the sample with links that are the first and only links of the parent firm, and (iii) the sample that excludes links of parents of 'unknown' sector. Second, with respect to features of the subsidiaries, we consider (iv) the sample that excludes links of subsidiaries classified as financial sector, holding company or unknown sector and (v) the sample that excludes links with subsidiaries that do not report basic balance sheet data, e.g. total assets or operating revenue. The local projection estimates for (i)-(iii) are shown in Figure A.4, and the estimates from (iv)-(v) are shown in Figure A.5, both in the Appendix.

The first row of A.4 shows the estimates based on the full sample of new links, which is not limited to first links but also includes any new link formed by a parent firm. As a result, at the time of incorporation, parents may already have links to other subsidiaries, meaning that this sample also includes firms that are already multinational. Consequently, this specification does not capture only the extensive margin of the decision to become a multinational. The qualitative pattern of the baseline results remains unchanged, although the magnitude of the effects is marginally larger. In particular, for the UK-EU case, the maximum correlation between the BUI and new subsidiary links still occurs three months after the increase in the BUI, but it is larger and closer to 6. The cumulative effect after six months amounts to a 3% increase relative to the average six-month number of 912 new subsidiaries in the period of estima-

tion, per one percentage point of BUI. For the EU-UK case, the contemporaneous (month 0) correlation is almost 1.5 times larger, and it displays a clear declining pattern over the six-month horizon. This indicates that the baseline findings are not driven by the restriction to first-time multinational activity and that heightened uncertainty affects both the extensive margin of becoming a multinational and the subsequent expansion decisions of firms that are already internationally active.

The second row of Figure A.4 reports the local projection estimates for the subsample of parents with a first and only one subsidiary in the period 2011-2021. Our baseline results are also robust to this variation and in line with the corresponding DiD robustness exercise, as confirmed by the responses in the figure.

As an additional robustness check, we again verify that the results are not driven by parent firms whose sector is classified as '*unknown*'. The last row of Figure A.4 reports the local projection estimates when restricting the sample to parents with a known sector classification. The results remain qualitatively unchanged for both the UK-EU and EU-UK samples. In particular, we continue to observe a delayed and persistent response in EU subsidiary links formed by UK parents, alongside a flatter response in UK subsidiary links formed by EU parents. Quantitatively, the effects also remain in both directions, although the peak response in the UK-EU sample is slightly attenuated.

Finally, to ensure that our results are not driven by parents not being the ultimate owner of the subsidiaries, we restrict our sample to the links where parents are also listed as the global ultimate owner (GUO) of 25% and of 50% of the subsidiary. This excludes, for instance, a link between a British parent firm with an EU subsidiary whose eventual majority owner is any firm other than the same British parent, and thus controls for indirect ownership. We note that the GUOs of 25% are not a perfect subset of the GUOs of 50%, and that our imposed restriction is the most conservative one. This reduces our UK-EU and EU-UK samples to a third of the baseline, with 6,087 distinct links for UK-EU, and 4,762 distinct links for EU-UK (Table A.1). Figure A.4 shows that our results remain qualitatively unchanged.

We now turn to the subsidiary characteristics exploiting their sector and their balance sheet figures. A first concern is that a large share of newly created subsidiaries may belong to the financial sector. In particular, the EU-UK results could potentially be driven by EU parent firms seeking to maintain access to British financial services. To address this, we repeat the local projection estimation after excluding all links for which the subsidiary operates in the financial sector or the sector is unknown, or classified as a holding company. The resulting samples therefore include only links in which both parent firms and subsidiaries are non-financials and represent 62.1% of the baseline UK-EU sample and 68.8% of the EU-UK baseline sample. The first row of Figure A.5 shows that the baseline results are virtually identical when using this restricted sample, indicating that access to financial markets is not driving the results.

A second concern is whether the subsidiaries established by parent companies are entities with no real business activities, and exist only on paper. Our dataset does not allow us to strictly identify such potential '*shell*' subsidiaries, because both in the UK and EU it was not a legal requirement for small entities to report detailed balance sheet data during the time period of our samples. Therefore, lack of balance sheet data entries can mean either that the company is not reporting it or that the company has no business activity. We can nevertheless restrict the sample to contain first subsidiaries that report some business activity in the period that they are linked to a parent firm, and consider this sample a proxy for excluding such paper-based companies. Specifically, we require that each first subsidiary in our samples

reports some business activity in the form of total assets or operating revenue, at least once between the one year before and one year after the link with the parent firm. With this refinement, the number of links is approximately halved for the the UK-EU sample and drops by one third for the EU-UK. Despite that, as shown in the second row of Figure A.5, the local projection estimates continue to exhibit the same qualitative features as in the baseline, albeit at lower levels for both UK-EU and EU-UK samples. This evidence supports the view that the response of British and European firms that established subsidiaries is attributed to fear of losing access to foreign trade markets rather than tax evasion or other illicit activity.

5.3 Response to Brexit uncertainty at sectoral level. Our data further allow us to explore heterogeneity in firms' responses by focusing on the sector of the parent company. Table A.10 shows the breakdown of sectors for the UK-EU and EU-UK samples, excluding financial firms and holding companies (using the same classification procedure as with the other samples). Figures 7a-7b report the local projection estimates of the effects of BUI_t for the UK-EU and EU-UK samples, respectively, disaggregated by the sector of the parent firm that established a first subsidiary link between 2016 and 2021.[17] This sectoral breakdown provides additional insight into which types of firms are primarily driving the aggregate patterns documented in the baseline results.

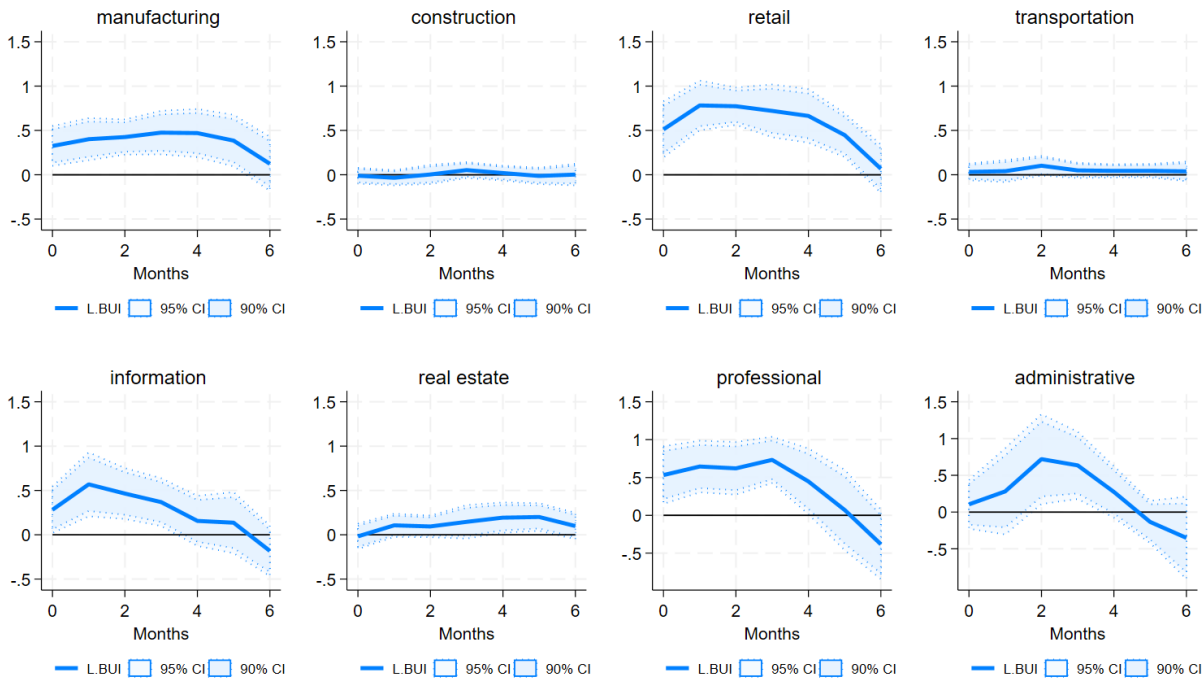
For the UK-EU sample, five sectors account for the bulk of the overall effect: manufacturing, retail, information, professional services, and administrative activities. Among these, the responses of the retail, professional, and information sectors are particularly pronounced, both in terms of magnitude and persistence. These sectors exhibit a clear and sustained increase in subsidiary formation after increased uncertainty, suggesting that firms operating in activities that rely heavily on cross-border market access, client proximity, and regulatory continuity were especially responsive. This pattern is consistent with anecdotal evidence surrounding Brexit-related corporate restructuring and is in line with existing findings for service-sector firms, as documented by Breinlich and Magli (2025).

In contrast, the sectoral responses in the EU-UK sample are generally more muted, mirroring the weaker aggregate effects observed in the baseline analysis. Nevertheless, two sectors stand out: professional services and information. The responses in these sectors suggest that EU-based firms in knowledge-intensive and client-facing activities were more likely to establish a UK presence in response to increased uncertainty. This is particularly indicative of professional service providers, such as legal, consulting, and advisory firms, for whom maintaining direct access to foreign markets and clients is critical. Overall, the sectoral evidence supports the interpretation that these location decisions were shaped by firms' efforts to preserve operational continuity and access to key markets in activities that are highly sensitive to regulatory and institutional changes.

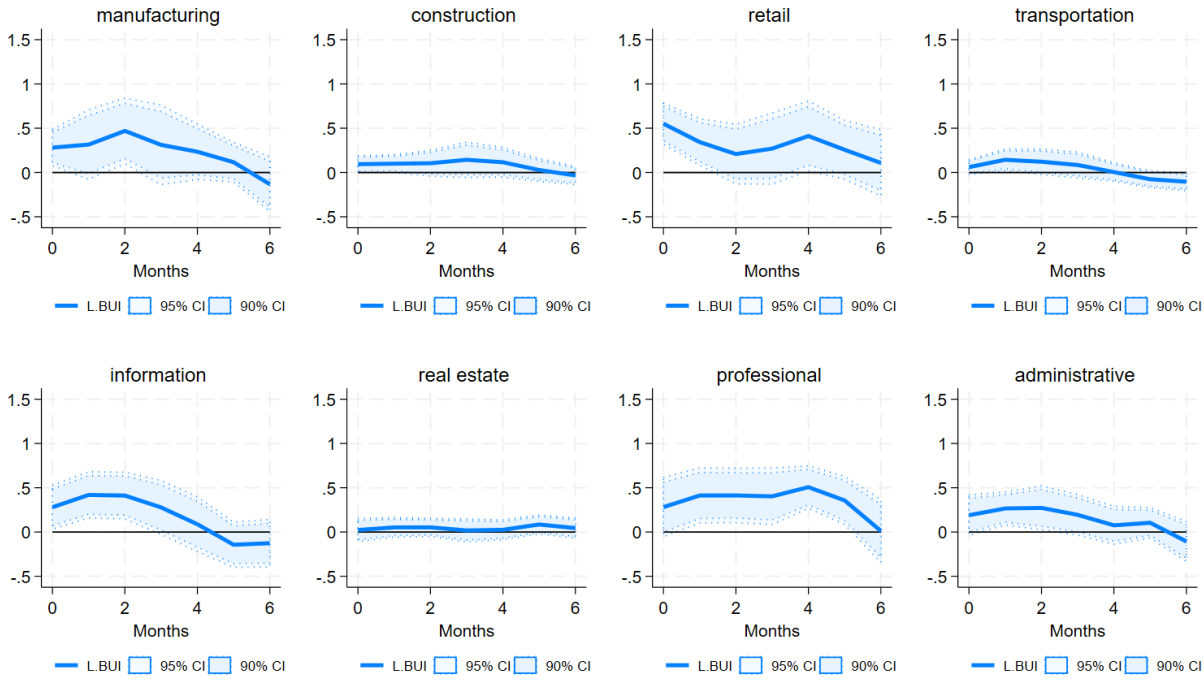
[17] We only present the estimated effects for the eight largest sectors in our samples, after excluding financial sector and holding companies. Using NACE classification rev. 2, these are *manufacturing* (C), *construction* (F), *wholesale and retail trade* (G), *transportation and storage* (H), *information and communications* (I), *real estate activities* (L), *professional, scientific and technical activities* (M), and *administrative and support services* (N).

FIGURE 7: Dynamic Responses of first new Links to Uncertainty by Parent Sector

(A) UK → EU



(B) EU → UK



Note: Top panel (A): UK-EU sample and bottom (B): EU-UK sample. LPs of first new links by parent sector on (i) BUI_{t-1}, (ii) CUI_{t-1}, (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical, Office of National Statistics UK, Bank of International Settlements* and *Eurostat.*

6. A Model of Subsidiary Formation under Expected and Uncertain Costs

The empirical results present a puzzle. After the referendum, more UK firms established their first subsidiary in the EU, and these new parents had weaker pre-incorporation performance. Fewer EU firms established their first UK subsidiary, and the new parents had stronger pre-incorporation performance. Yet, during the post-referendum period, increases in the Brexit Uncertainty Index (BUI) were associated with more first subsidiaries in *both* directions, particularly among small firms. The first two concern pre- versus post-referendum changes; the third concerns fluctuations within the post-referendum period. The mechanisms accounting for them need not be identical.

Our starting point is the proximity–concentration framework of Brainard (1997) and Helpman et al. (2004), in which heterogeneous firms sort between serving a foreign market from abroad and establishing a local affiliate. Three related literatures clarify why uncertainty may change this choice. Ramondo et al. (2013) show how volatility can favour exporting relative to affiliate sales through the flexibility of cross-border provision. Related work on risk aversion and international trade shows how uncertainty shapes firms' sourcing, exporting and geographical diversification decisions (Gervais, 2018; De Sousa et al., 2020; Esposito, 2022). In models of trade-policy uncertainty with sunk costs of establishing a subsidiary, waiting until policy is resolved has option value and can delay investment (Handley and Limão, 2015, 2017).

Our framework brings these considerations into a common organisational decision. Its distinctive feature is that postponement does not eliminate risk: the firm continues to serve the destination from abroad, so waiting preserves flexibility while maintaining exposure to uncertain market-access costs. We derive a decomposition separating expected cost changes, the convexity of cross border profits, the insurance value of local presence, and the net gains from future switching. This yields conditions under which greater uncertainty brings subsidiary formation forward, despite sunk organisational costs, and links that response to the selection of firms establishing their first subsidiary. The contribution is thus not a new theory of risk aversion or of real options in isolation, but a characterisation of their competing effects on the timing and composition of foreign subsidiary formation. Formal propositions and proofs are provided in Appendix C and the discussion here focuses on the underlying economics in relation to the empirical evidence.

6.1 Firms' organisational choices. We extend the heterogeneous-firm approach to exporting and multinational production in Helpman et al. (2004) to allow firms to choose their organisational form before the future costs of serving a market are known. There are two time periods $t = 0, 1$, and a continuum of monopolistically competitive firms that differ in their ability to generate profits $\phi > 0$, drawn from a distribution G. Capability summarises the scale and profitability of a firm's activities in a given destination: at a given market-access cost, a more capable firm earns higher variable profits there. A firm can generate operating profits by serving a foreign destination market or from other unrelated activities that are independent of how the foreign market is served. We denote the latter component of profits with $\pi_D(\phi)$. A firm can serve a foreign destination market either from its home country by exporting (mode X) or through a foreign (local) subsidiary in the destination country (mode I). The distinction is between the costs of serving the market across a border and those of maintaining a local entity, rather than between

particular forms of foreign production. Foreign demand comes from CES preferences with elasticity of substitution $\sigma > 1$.

We define the following functions to represent total operating profits from either serving the destination market by exporting Π_X or through a foreign subsidiary Π_I:

$$\Pi_X(\phi;\tau) = \pi_D(\phi) + \tau^{1-\sigma}\phi, \qquad \Pi_I(\phi;F_I) = \pi_D(\phi) + \alpha\phi - F_I. \tag{7}$$

The function Π_X is parametrized by the variable (iceberg) cost $\tau \geq 1$, i.e. the proportional cost of serving the destination from abroad. Depending on the firm's activities, it can represent transport or tariff costs, or the variable component of regulatory and market-access frictions. The function Π_I is parametrized by $F_I > 0$, i.e. a recurring fixed operating cost F_I and a parameter $\alpha > 0$ that allows for differences in variable profitability between local and cross-border provision. We assume $\alpha > \tau^{1-\sigma}$, so that the subsidiary offers higher variable profit but requires a fixed outlay and the gain from local presence increases with capability. This is the conventional proximity-concentration sorting force: more capable firms are better placed to cover the cost of a subsidiary (Brainard, 1997; Helpman et al., 2004).

The profits $\pi_D(\phi)$ do not depend on the organisational mode used to serve the foreign destination but will matter under uncertainty. A firm with a small share of foreign destination profits to overall profits can absorb market-access shocks more easily than an otherwise similar firm whose profits depend heavily on that destination. Keeping this background component in total profits allows exposure to affect the incentive to establish a subsidiary, without changing the basic sorting mechanism under known costs.

Finally, we also assume that in any period, changing the way the foreign destination is served involves a one-off switching cost $\kappa > 0$, which is sunk once paid, and has to be paid every time the firm changes mode. The size of κ therefore parametrizes how costly it is to postpone a decision to switch, or to undo it; it is distinct from F_I, which is a recurring fixed cost paid in every period in which a foreign subsidiary operates.

The timing captures the period of anticipation following the referendum. In period 0, the existing arrangements remain in place and the costs $(\tau_0, F_{I,0})$ are known. At the start of period 0, all firms are in mode X, i.e. they have no subsidiaries in the foreign destination. Each firm decides whether to continue serving the destination from abroad or to establish local presence immediately. Setting up or closing down a foreign subsidiary entails a sunk organisational cost $\kappa > 0$, separate from the recurring cost F_I. This may include incorporation, regulatory approvals and the reorganisation of customer or supplier relationships, or relocation and closing down costs.[18]

In period 1, the new trading environment is realised, but the costs are uncertain, and thus stochastic, from the perspective of period 0, and defined as $\tau_1 = \mu_\tau + \varepsilon_\tau$ and $F_{I,1} = \mu_F + \varepsilon_F$. The noise terms have zero means and variances σ_τ^2 and σ_F^2. The referendum can therefore alter both expected future costs and the uncertainty surrounding them, even though contemporaneous period 0 costs remain unchanged. Once the new costs are observed, firms may retain their chosen mode or change it by paying κ again. A firm that postpones subsidiary formation preserves the ability to establish a subsidiary later, but continues to

[18] We note that the initial mode X is a modelling convention, given that our data set tracks subsidiary links but not whether the firms without a destination subsidiary were previously exporting there.

bear the risks of cross-border provision in the meantime. Conversely, a firm that establishes a subsidiary immediately gains local presence before the new regime is known, while retaining the costly possibility of reversing the decision. The switching cost is thus what makes initial organisational choices consequential for future payoffs.

To represent the cost of uncertain profits, we use the certainty equivalent $CE_0(Y) = u^{-1}(\mathbb{E}_0 u(Y))$, where $u(Y) = Y^{1-\gamma}/(1-\gamma)$ for $\gamma \neq 1$, $u(Y) = \log Y$ for $\gamma = 1$, and $\gamma \geq 0$ measures relative risk aversion. A certainty equivalent expresses the valuation of future profit in monetary units, so that it can be compared with current operating profit and switching costs. In particular, a certain payoff has a certainty equivalent equal to that payoff: risk aversion affects the organisational decision only when future profits are uncertain. We assume that total profits remain positive in the relevant states.

Next we proceed with two definitions that are central for the model analysis. The first is the *gain* in period 0 from establishing a subsidiary immediately, rather than remaining in mode X, defined as

$$D_0(\phi) = (\alpha - \tau_0^{1-\sigma})\phi - F_{I,0} - \kappa + \delta\big[CE_0(\Pi_{I,1}) - CE_0(\Pi_{X,1})\big], \tag{8}$$

where $\Pi_{X,1}$ and $\Pi_{I,1}$ denote period 1 profits conditional on starting that period in mode X and I, respectively, after the firm optimally decides whether to switch. The first part of the gain $Z_0(\phi) \equiv (\alpha - \tau_0^{1-\sigma})\phi - F_{I,0}$ is the *current operating advantage* of local presence net of the sunk cost of establishing a subsidiary. The last part $CE_0(\Pi_{I,1}) - CE_0(\Pi_{X,1})$, weighted by $\delta \in (0,1]$, is its discounted *continuation advantage*. The latter incorporates uncertainty about future costs and the opportunity to change modes once they are realised. Appendix C sets out the switching problem and shows that the decision is described by a single cutoff when future costs are known or firms are risk neutral; under risk aversion the sorting property holds for small risks and is verified numerically in every configuration considered below. The second definition is that of a *subsidiary-formation cutoff* ϕ_I^*, defined by $D_0(\phi_I^*) = 0$. Under the sorting condition, firms with capability above the cutoff establish subsidiaries and those with capability below it do not. The mass of new parents is therefore $1 - G(\phi_I^*)$. A fall in the cutoff raises the number of new parents and draws in firms with lower capability, while a rise in the cutoff reduces subsidiary formation and strengthens selection. We interpret observed first-subsidiary formation as decisions by successive cohorts of firms facing the prevailing beliefs about future costs. The model does not require all firms to make this decision on the same calendar date. The common cutoff mechanism is what allows us to relate changes in subsidiary counts to the pre-incorporation firm characteristics examined in Section 3.

6.2 The determinants of subsidiary formation. The decision of a firm to establish a subsidiary before the future trading regime is known depends on the gain $D_0(\phi)$ which in turn should depend on (i) the expected profitability of local presence with a subsidiary, (ii) the general uncertainty that affects both organisational modes, and (iii) the value of being able to revise the choice later. In what follows, we explain how the gain can indeed be decomposed in components that match these. We will use this decomposition to analyse changes in expected costs and changes in uncertainty.

The gain from establishing a new subsidiary in $t = 0$ can be expressed as (see Appendix C for formal

derivation):

$$D_0(\phi) = Z_0(\phi) - \kappa + \delta\Big(\bar{Z}_1(\phi) - J(\phi) + P(\phi) - W(\phi)\Big) \tag{9}$$

where the continuation advantage can be exactly decomposed in four terms. (i) The first term $\bar{Z}_1(\phi) \equiv (\alpha - \mu_\tau^{1-\sigma})\phi - \mu_F$ measures the advantage of local presence at *expected future costs*. An increase in expected cross-border costs raises this advantage, whereas higher expected subsidiary operating costs reduce it. (ii) The second term $J(\phi) = \big(\mathbb{E}_0(\tau_1^{1-\sigma}) - \mu_\tau^{1-\sigma}\big)\phi \geq 0$ captures the *convexity gain* from uncertain proportional trade costs accruing to firms that continue to serve the destination from abroad, due to Jensen's inequality, and works against new subsidiary formation. (iii) The third term $P(\phi) \equiv RP(\Pi_{X,1}) - RP(\Pi_{I,1})$ compares the risk premia under the two modes of operation, where $RP(Y) \equiv \mathbb{E}_0 Y - CE_0(Y)$: it favours local presence when a subsidiary provides insurance against uncertain cross-border costs, i.e. *precautionary subsidiary formation*, but can be negative if subsidiary profits are riskier. (iv) Finally, the term $W(\phi)$ is the *option value of waiting* to establish a subsidiary later net of the value of reversing an earlier establishment decision, capturing the flexibility afforded by future switching. We note that the identity (9) holds whether or not firms change organisational mode in period 1.

The decomposition in these four terms brings together the proximity-concentration mechanism of Helpman et al. (2004), the flexibility of exporting under uncertainty emphasized by Ramondo et al. (2013), the consequences of uncertain profits for the exporting-versus-affiliate choice, and the option value of waiting with sunk costs (Handley and Limão, 2015, 2017). The decomposition combines them within the same choice rather than assigning one mechanism to the entire response. Waiting retains flexibility, but does not remove the uncertainty faced by a firm continuing to serve its destination as an exporter. In what follows, we now focus on two key components, i.e. the expected cost changes (subsection 6.2.1) and the competing effects of uncertainty (subsection 6.2.2).

6.2.1 Expected Brexit costs: subsidiary formation and selection. To understand the role of changes in expected future costs, μ_τ and μ_F, we assume first that their variances are zero. Such changes are in the first moments of the post-Brexit cost distributions, and not uncertainty shocks. A rise in the expected cost of serving a market from abroad increases the relative profitability of local presence: firms can establish subsidiaries before the new regime takes effect, preserving market access and avoiding the higher future friction. Conversely, a rise in the expected recurring cost of operating a subsidiary makes local presence less attractive. When the marginal firm does not plan to change organisational mode in period 1 (we call this *inaction*), the cutoff falls if the expected loss from serving the destination from abroad exceeds the expected increase in subsidiary operating costs, i.e. whenever

$$\big(\tau_0^{1-\sigma} - \mu_\tau\big)\phi_{I,0}^* > \mu_F - F_{I,0}, \tag{10}$$

where $\phi_{I,0}^*$ is the initial cutoff, evaluated before the change in expectations. The inequality compares the two cost changes at the same initially marginal firm. Outside this inaction region the marginal firm plans to switch at $t = 1$, and the cutoff no longer responds to expected future costs (see Appendix C.3).

This condition explains why the patterns of subsidiary formation and selection differ across direc-

tions. For UK firms considering an EU subsidiary, the referendum *raised* anticipated costs of accessing the Single Market from outside. If these costs increased more than the cost of operating an EU subsidiary, the cutoff fell: more firms established subsidiaries, and the additional new parents were weaker (lower capability ϕ). For EU firms considering a UK subsidiary, greater expected cross-border friction alone would also encourage subsidiary formation. However, the observed combination of *fewer* and *stronger* new parents instead requires a sufficiently large deterioration in the expected relative profitability of EU subsidiaries in the UK.

This extension of the familiar productivity sorting logic (Helpman et al., 2004) is useful for interpreting subsidiary formation and selection jointly: the relevant margin is a change in the cutoff, not necessarily an improvement in the performance of firms that establish a subsidiary. We note that this explanation of post referendum subsidiary formation and selection does not require an increase in uncertainty. Expected cost changes should therefore be distinguished from changes in uncertainty when interpreting the empirical evidence.

6.2.2 Uncertainty: convexity and precautionary subsidiary formation. Next, we isolate two uncertainty mechanisms when the marginal firm is unlikely to change its organisational mode after future costs are realised, $W = 0$. Convexity of export profits in proportional trade costs favours remaining in mode X, since firms can adjust sales when costs are realised, as in Ramondo et al. (2013). Risk aversion works in the opposite direction: local presence via subsidiary formation can insure against market-access risk, although uncertainty about subsidiary operating costs makes local presence itself less attractive.[19]

To understand the precautionary mechanism, first consider the marginal firm for which switching modes after costs are realised is sufficiently unlikely to affect the local small-risk comparison. With only uncertain proportional trade costs, the insurance motive dominates the convexity effect and $\frac{\partial \phi_I^*}{\partial \sigma_\tau} < 0$ when

$$\gamma\, s(\phi_I^*; \mu_\tau) > \frac{\sigma}{\sigma - 1}, \tag{11}$$

where

$$s(\phi; \mu_\tau) \equiv \frac{\mu_\tau^{1-\sigma} \phi}{\pi_D(\phi) + \mu_\tau^{1-\sigma} \phi}. \tag{12}$$

Here ϕ_I^* is the subsidiary-formation cutoff defined by $D_0(\phi_I^*) = 0$, evaluated at the zero-uncertainty benchmark so that condition (11) is therefore assessed at the marginal firm. The share s measures the importance of the foreign destination market in total firm profits. A firm that is more reliant on the foreign market (larger s), or one for which profit risk is more costly (larger risk aversion γ), has a stronger incentive to insure itself through local presence. We can show also that uncertainty about subsidiary costs produces the opposite effect: it increases the relative attractiveness of serving the destination from abroad, $\frac{\partial \phi_I^*}{\partial \sigma_{F_I}} > 0$.

[19] In this local inaction benchmark, the option term W is zero; its implications are considered in the numerical analysis in Section 6.2.3.

We now allow uncertainty about cross-border market access to affect both proportional trade costs and recurring fixed costs, such as certification, customs documentation or regulatory compliance. The distinction matters because the two types of costs affect firms differently. An increase in proportional trade costs reduces profits in proportion to the scale of foreign operations, whereas an increase in fixed compliance costs reduces profits by the same amount regardless of firm capability. Both expose firms to uncertainty, but only proportional trade costs generate the convexity benefit of serving the market from abroad.

To capture this distinction, suppose that the two costs are driven by the same underlying shock (ε_τ). In addition to the proportional trade cost $\tau_1 = \mu_\tau + \varepsilon_\tau$, a firm serving the destination from abroad faces a recurring fixed cost $F_{X,1} = \mu_X + k\varepsilon_\tau$, where $k \geq 0$ measures the sensitivity of fixed compliance costs to the shock. This cost is absent in period 0 and in the benchmark model. Export profits and the decomposition in decomposition (9) are therefore adjusted to include this additional cost, in particular, $\tilde{Z}_1$ includes μ_X (see Appendix C.4).

We define $\lambda(\phi)$ as the fraction of a firm's exposure to the common shock attributable to fixed compliance costs. Since the sensitivity of export profits to proportional trade costs is $(\sigma-1)\mu_\tau^{-\sigma}\phi$, this fraction is $\lambda(\phi) = \frac{k}{(\sigma-1)\mu_\tau^{-\sigma}\phi + k}$. For a given k, $\lambda(\phi)$ decreases with firm capability: fixed compliance costs account for a larger share of the exposure of smaller firms to market-access uncertainty. The condition for the insurance benefit of local presence to dominate the convexity effect then become:

$$\gamma s(\phi_I^*; \mu_\tau) > \left(1 - \lambda(\phi_I^*)\right)^2 \frac{\sigma}{\sigma - 1}. \tag{13}$$

A fixed component increases profit risk without creating the corresponding convexity benefit of exporting, making precautionary subsidiary formation easier to obtain. This distinction is pertinent to Brexit because compliance, certification and administrative requirements do not necessarily scale with sales. It also shows why the prediction for uncertainty differs from models in which the relevant risk is primarily in variable production or trade costs (Ramondo et al., 2013). It can also be particularly important for small firms, for which fixed frictions represent a larger fraction of total profit risk. Smaller firms may additionally be more dependent on one destination and face higher effective costs of adverse profit outcomes. These are distinct mechanisms consistent with the empirical findings related to size heterogeneity but the empirical estimates do not identify them separately.

The conditions (11) and (13) provide a comparative static result for the marginal firm: they characterise when the insurance or precautionary motive overturns the convexity effect for small risks under inaction, not at arbitrary levels of risk. At larger risks, the option to wait term W can become important because firms may change their organisational form once costs are realised. We turn to this additional mechanism in in the next Subsection 6.2.3, where all four terms of the decomposition operate together.

6.2.3 Numerical illustration. The analytical results so far identify when uncertainty may encourage subsidiary formation, but they do not establish the size of the response once firms can switch organisational form. To make progress on this front, we next solve the full model numerically, allowing the convexity, precautionary and option terms in the decomposition to operate together. This allows us to

illustrate how reasonable configurations of the model can reproduce the main empirical patterns we observe, and to identify which mechanism each of them rests on.[20] Appendix C.6 provides the details on the numerical solution method.

For the numerical experiments, we set the CES elasticity to $\sigma = 4$ and risk aversion to $\gamma = 4$. Since the two periods in the model represent two economic regimes rather than calendar periods of equal duration, we abstract from discounting and give future firm profits the same weight as current profits by setting $\delta = 1$. As in Helpman et al. (2004), we use a Pareto distribution of capability with shape parameter 1.06 to translate movements in the subsidiary formation cutoff into changes in the number of new parents.

To discipline the change in cross-border costs, we draw on the soft and hard Brexit scenarios in Dhingra et al. (2017), which imply increases in non-tariff trade barriers of approximately 2.8% and 8.3%, respectively. Giving these scenarios equal probabilities suggests an increase in the expected increase of about 5.5% and a dispersion of $\sigma_\tau = 0.031$ around the post-referendum trade-cost level. These figures translate the spread between the scenarios into a two moment description of uncertainty. The pre-referendum proportional trade cost is set at $\tau_0 = 1.10$, i.e., a 10% avoidable friction inside the Single Market.

An important determinant of the precautionary motive is the share of a firm's total profits associated with the foreign market under consideration, s. For UK parents establishing EU subsidiaries, we set $s = 0.455$, the average of the annual EU shares of aggregate UK exports of goods and services over the period 2011-2015, computed from the trade statistics of the *Office for National Statistics*. The share in 2015 alone was 44%. For EU parents establishing UK subsidiaries, we set $s = 0.15$, provided by *Eurostat* in 2015, to represent a smaller exposure to the UK market. Aggregate export shares are proxies for firm-level profit exposure, which we do not observe.[21] We assume in the benchmark that half of the Brexit related cross-border friction is a recurring fixed cost ($\lambda = 0.5$), to capture administrative and regulatory requirements that need not rise with sales. [22]

The pre-referendum recurring cost of a subsidiary, $F_{I,0}$, is chosen so that the marginal firm is indifferent between the two modes before the referendum, after normalising its capability to $\phi = 1$. Together with a switching cost κ equivalent to approximately two years of that firm's pre-referendum annual export profit, this gives $F_{I,0} = 0.080$, or about 8% of its subsidiary variable profit. For UK parents establishing EU subsidiaries, we hold the expected future subsidiary cost at its initial level, $\mu_F = F_{I,0}$, and set subsidiary cost uncertainty to zero ($\sigma_F = 0$). This benchmark reflects the fact that an EU subsidiary remains within the EU regulatory and legal framework after Brexit: the main new friction for a UK firm is assumed to arise from serving the EU market from outside it (as an exporter), rather than from operating an established EU entity. For EU parents establishing UK subsidiaries, by contrast, Brexit may alter the costs of operating locally in the UK through changes in regulation, administration, or access to personnel. We first show the unchanged-cost benchmark, then raise μ_F by approximately 0.14, i.e. the change required to reproduce

[20] In what follows we do not attempt to either estimate or calibrate the model, but we simply look at its predictions when we employ reasonable values and ranges for the parameters of interest.

[21] Figure 9 shows the sensitivity of the results to changes in s.

[22] This share is imposed for the pre-referendum marginal firm. Because the post-referendum marginal firm is less capable, the fixed share of its own profit risk is $\lambda(\phi_I^*) \approx 0.56$, the value at which condition (13) is evaluated. This assumption ensures that a reasonably low firm risk aversion coefficient is sufficient for condition (13) to be satisfied.

TABLE 5: Expected costs, uncertainty, and foreign subsidiary formation

	Contribution of each term					Response to
	$\bar{Z}_1$	J	P	W	Total	**uncertainty**
A. UK → EU ($s = 0.455$, $\sigma_\tau = 0.031$, $\mu_F = F_{I,0}$, $\sigma_F = 0$)						
$\lambda = 0.5$	27.15	−0.15	1.14	−0.02	28.12	0.97
$\lambda = 0$	24.07	−0.57	0.78	0.00	24.27	0.21
B. From UK → EU to EU → UK: changing one assumption at a time ($\lambda = 0.5$)						
Lower share: $s = 0.15$	27.15	−0.15	0.37	−0.01	27.35	0.21
Also $\mu_F = F_{I,0} + 0.14$	−4.40	−0.11	0.21	0.00	−4.30	0.10
Also $\sigma_F = 0.02$	−4.40	−0.11	0.18	0.00	−4.33	0.07

Notes: Changes in the number of fist time new parents relative to the pre-referendum level, in percent. Common parameters: $\sigma = 4$, $\gamma = 4$, $\kappa \approx 2$ years, and a 5.5% expected increase in cross-border trade costs in both directions. The $\bar{Z}_1$ column reports the change in subsidiary formation with known costs at the expected post-referendum values; J, P, and W give the incremental changes from adding each term of (9) sequentially. Total is the sum for the four first columns. Each column is the difference between two recomputed cutoffs converted into numbers of firms. The last column measures the additional response when future costs become uncertain at the same expected values (Total minus $\bar{Z}_1$). Panel A assumes $\mu_F = F_{I,0}$ and $\sigma_F = 0$. Panel B starts from the first row of Panel A and changes s, μ_F, and σ_F sequentially, for given $\lambda = 0.5$.

the observed 4.4% decline in first subsidiary formation under known future costs. We also set $\sigma_F = 0.02$, to correspond to a standard deviation of around 2% of the marginal firm's subsidiary variable profit.

Table 5 presents two related numerical exercises. In each row, the $\bar{Z}_1$ column measures the change in the number of first-time parents when the expected post-referendum costs replace their pre-referendum values, but future costs are treated as known. The remaining columns add trade-cost uncertainty: column J shows the incremental effect due to convexity under risk neutrality, column P shows the further contribution of risk aversion, and column W reports the additional effect of allowing firms to change organisational mode in period 1. These are sequential contributions, in percentage points of pre-referendum subsidiary formation, each obtained by recomputing the cutoff with one more term active; they are not the terms of (9) themselves, which are measured in units of profit for a given firm. The last column isolates the overall response to uncertainty, holding the post-referendum means fixed. This distinction between the changes in expected costs and the additional effect of uncertainty is central to reading the table.

Starting with the results in the top panel of the table, i.e. with the assumption that Brexit changes the cost of serving the EU from the UK, but not the recurring cost of operating an EU subsidiary: $\mu_F = F_{I,0}$ and $\sigma_F = 0$. Thus, the subsidiary operating cost is the same in both periods and is known with certainty; only cross-border provision is exposed to Brexit-related cost risk. The first row assumes that half of the Brexit-related cross-border friction is fixed ($\lambda = 0.5$). At the new expected trade costs, but without uncertainty, subsidiary formation increases by 27%. The corresponding fall in the cutoff allows additional, less capable UK firms to form foreign subsidiaries. Introducing uncertainty adds −0.15 percentage points through the convexity of export profits, +1.14 percentatge points through the insurance value of local presence for risk-averse firms, and −0.02 percentage points through the option to wait. The net response

to uncertainty is therefore +0.97 percentage points, bringing the total increase in subsidiary formation to 28%. This exercise illustrates how higher expected cross-border costs can explain the increase in subsidiary formation and weaker firm selection, while uncertainty further encourages subsidiary formation through a distinct precautionary FDI mechanism.

The second row of the top panel of Table 5 changes one assumption: all Brexit related cross-border frictions are now proportional ($\lambda = 0$), rather than half fixed. We retain $s = 0.455$, the same assumed expected aggregate loss from Brexit, the same overall scale of cross-border profit risk, and $\mu_F = F_{I,0}$ with $\sigma_F = 0$. Under this alternative composition of the friction, the known cost increase in subsidiary formation is 24%, and the convexity contribution becomes more negative (−0.57 points) because a larger proportion of the shock operates through the convex export-profit function. The precautionary contribution is still positive (+0.78 points) but smaller relative to convexity, leaving an uncertainty response of only +0.21 points. The comparison shows why the composition of the friction matters: fixed compliance costs generate profit risk without the same offsetting convexity benefit of exporting.

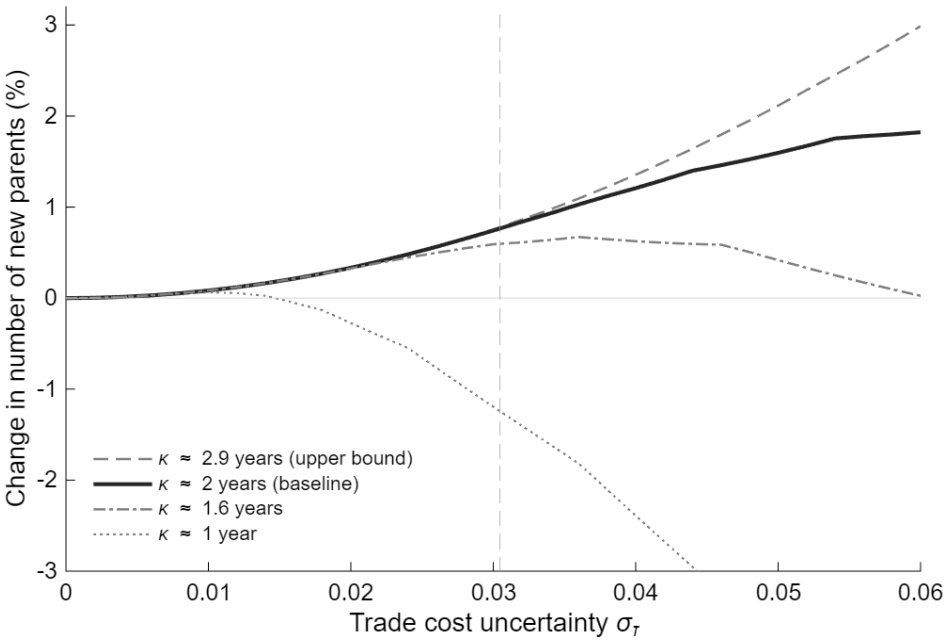
In the lower panel of the table, we start from the UK parent benchmark that was used in the first row of the top panel, with half of the Brexit-related friction fixed ($\lambda = 0.5$), and progressively introduce the three assumptions that distinguish EU parents establishing UK subsidiaries. This sequence serves two purposes: it identifies which assumptions account for the opposite changes in the *level* of first subsidiary formation after the referendum, and explains why the additional response to uncertainty is smaller for EU parents. Each row changes only the parameter indicated, retaining the assumptions introduced in the previous rows.

The first row lowers the destination's share of the firm's profits from $s = 0.455$ to $s = 0.15$, keeping the expected recurring subsidiary cost unchanged ($\mu_F = F_{I,0}$) and certain ($\sigma_F = 0$). With the same expected rise in cross-border costs, establishing a UK subsidiary would still become more attractive to EU parents: under known future costs, first subsidiary formation increases by 27%, just as for UK parents. However, the additional response to uncertainty falls from 0.97 to 0.21 percentage points, mostly because smaller destination market exposure reduces the insurance benefit of a foreign subsidiary. Lower exposure helps explain the weaker EU uncertainty response, but cannot explain the observed reduction in EU-to-UK subsidiary formation after the referendum.

The second row therefore increases the expected recurring cost of operating a UK subsidiary, to $\mu_F = F_{I,0} + 0.14$, while keeping $\sigma_F = 0$. This increase is the one that reproduces the observed 4.4% reduction in first subsidiary formation when future costs are known. The deterioration in the expected payoff from the foreign subsidiary now more than offsets the incentive to avoid higher cross-border costs. The subsidiary formation cutoff rises, implying fewer but more capable new parents. At the same time, the response to trade cost uncertainty falls further, from 0.21 to 0.10 percentage points. Thus, the same configuration can generate declining subsidiary formation following the referendum and a small positive response to uncertainty around the new expected cost environment.

Finally, the third row introduces uncertainty about the recurring cost of a UK subsidiary, setting $\sigma_F = 0.02$ while leaving its mean and the other parameters unchanged. Unlike the first two rows, establishing a UK subsidiary now also exposes the parent to uncertainty about operating costs. This weakens the subsidiary's insurance role: the additional response to uncertainty declines from 0.10 to 0.07 percentage

FIGURE 8: Trade cost uncertainty, subsidiary formation and switching cost

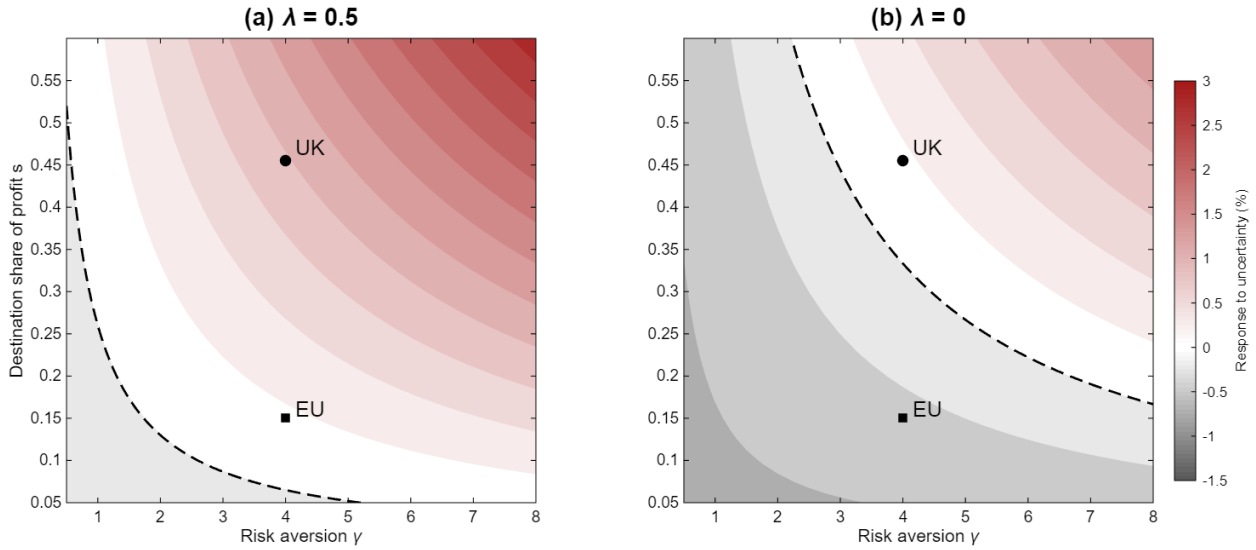


Notes: Change in the number of new parents when trade costs become uncertain, relative to known costs with the same means, for UK parents in the benchmark configuration ($s = 0.455$, $\gamma = 4$, $\lambda = 0.5$, $\sigma_F = 0$) and four switching costs κ: 1 year, 1.6 years, 2 years and 2.9 years of the pre-referendum marginal firm's annual export profit (2.9 years is the largest value consistent with a non-negative recurring subsidiary cost). The dashed vertical line marks the calibrated dispersion $\sigma_\tau = 0.031$.

points, giving a total change of −4.3% relative to the pre-referendum benchmark. The last row is the EU parent benchmark. Its positive uncertainty response is relatively fragile: it can turn negative under greater subsidiary cost risk, or entirely proportional cross-border frictions.

Next, we vary some key parameters of the model to understand when precautionary FDI dominates the option of waiting and vice versa. First, in Figure 8 we vary the cost of switching modes κ after uncertainty is resolved. When this cost is high, the insurance value of an early subsidiary dominates and the response of subsidiary formation to uncertainty is positive throughout the plotted range. With lower switching costs, the response peaks and eventually declines. With sufficiently inexpensive switching, uncertainty discourages early subsidiary formation. This is the economic reason that an increase in uncertainty can either accelerate or delay subsidiary formation: a firm that waits retains flexibility but still bears uncertain cross-border costs until it switches (Handley and Limão, 2015). Second, we vary both risk aversion γ and exposure s to capture changes in the left hand side of condition (13) that determine when the precautionary effect dominates Figure 9 maps the effects when varying these two parameters for two cases $\lambda = 0$ and $\lambda = 0.5$. A fixed component of cross border frictions substantially enlarges the region where uncertainty increases subsidiary formation. The numerical zero response boundary closely tracks the local analytical condition (13) evaluated at the marginal firm, so the condition is informative about the full solution at this switching cost. With half of the friction fixed, uncertainty raises subsidiary formation for every γ above one at the UK exposure and above two at the EU exposure. With a proportional friction, the region shrinks to high risk aversion and high exposure, which is why the EU benchmark then lies on the wrong side of the boundary. Precautionary FDI is therefore not confined to a single parameter

FIGURE 9: Trade cost uncertainty, subsidiary formation, risk aversion and exposure



Notes: Change in the number of new parents when trade costs become uncertain at the calibrated dispersion ($\sigma_\tau = 0.031$), in percent of pre-referendum subsidiary formation, all four terms active, baseline switching cost, $\sigma_F = 0$ and $\mu_F = F_{I,0}$. The dotted line is condition (13) evaluated at the marginal firm. The circle and the square mark the UK and EU benchmarks.

configuration, and the figure makes transparent which assumptions govern its strength.

6.3 Interpretation of the evidence through the lense of the model. Our proposed theoretical framework helps interpret the empirical results along two distinct margins. The pre- and post-referendum comparisons describe changes in subsidiary formation and the characteristics of new parent firms; the local projections describe how subsidiary formation varies with the Brexit Uncertainty Index (BUI) within the post-referendum period. The first margin can be explained by changes in expected costs, while the second is consistent with an additional precautionary response to uncertainty. These interpretations do not necessarily imply that the BUI identifies a pure change in the variance of future costs.

The increase in first UK-to-EU subsidiaries, accompanied by weaker pre-incorporation performance among new parents, is consistent with a decline in the subsidiary formation cutoff. The expected deterioration in cross-border market access makes an EU subsidiary relatively more attractive, provided it outweighs any increase in the cost of operating one. Conversely, fewer EU-to-UK subsidiaries and stronger new parents imply a higher cutoff, which requires a sufficiently large deterioration in the expected relative payoff from operating a UK subsidiary. The model does not identify which regulatory, administrative or other costs produced this change. Nor does it suggest that subsidiary formation itself caused weaker or stronger pre-incorporation performance: the outcome is a change in selection among firms establishing a subsidiary.

The positive association between the BUI and UK-to-EU subsidiary formation is consistent with precautionary subsidiary formation. A firm that postpones establishing an EU subsidiary continues to serve that market from abroad and therefore remains exposed to uncertain cross border costs. The smaller and shorter lived positive association for EU-to-UK subsidiaries is consistent with lower exposure to the UK market and possible uncertainty about the costs of operating locally. As Table 5 illustrates, a higher

expected cost of operating a UK subsidiary can reduce the overall number of new parents even when fluctuations in uncertainty are associated with some additional subsidiary formation. Thus, the opposite post-referendum changes in the two directions need not imply opposite responses to uncertainty.

The greater response among small firms is also compatible with the theoretical framework. Less diversified firms may depend more heavily on a particular destination, fixed cross border costs can account for a larger share of their profit risk and financing frictions can make adverse outcomes more costly. Independently, a lower subsidiary formation cutoff brings in firms previously just below it. The data do not separately identify these mechanisms. Responses in information and professional services are likewise consistent with the importance of regulatory access and local presence, but the model makes no sector specific quantitative predictions.

The BUI records the share of UK firms reporting Brexit as a major source of uncertainty. It may reflect news about expected future costs as well as their dispersion, and may be an imperfect measure of the uncertainty perceived by EU parents. Accordingly, the local projections do not isolate a mean-preserving spread or separately estimate the precautionary, convexity and option of waiting channels. The model instead establishes conditions under which their interaction is consistent with the signs and asymmetries in the data. Its broader implication is that postponing organisational investment does not eliminate risk: when firms remain exposed to uncertain market access costs and changing mode later is costly, greater uncertainty can encourage them to establish foreign subsidiaries sooner.

7. Conclusion

Brexit created a new margin of adjustment for firms: whether to establish a corporate presence across the border. Between the referendum and the end of 2021, first subsidiaries formation of UK firms increased by 20%, while first subsidiary formation by EU firms in the UK fell by 4.4%. UK firms establishing their first EU subsidiary after the referendum had weaker pre-incorporation performance than their comparison group, including around 8 percentage points lower turnover growth in the preceding year. EU firms establishing UK subsidiaries, by contrast, were stronger. Within the post-referendum period, increases in the Brexit Uncertainty Index were associated with more subsidiaries in both directions: roughly 23 additional first EU subsidiaries per percentage point of the index over six months for UK parents, around half that number for EU parents. About two-thirds of the UK response came from small firms, alongside suggestive evidence of declining domestic subsidiary formation. We find no comparable relationship for UK subsidiaries in Norway and Switzerland or for EU firms establishing subsidiaries elsewhere in the EU.

The model separates two forces that the empirical evidence captures together. Changes in expected costs can account for the opposite movements in subsidiary formation and selection: the subsidiary formation cutoff falls for UK firms facing higher expected costs of serving the EU from abroad, while a sufficiently large increase in the expected cost of operating a UK subsidiary raises the cutoff for EU firms. Uncertainty introduces an additional mechanism. A firm that postpones establishing a subsidiary continues to serve the destination from abroad, so waiting preserves flexibility without eliminating exposure to uncertain cross-border costs. When a meaningful share of these frictions takes the form of fixed compliance costs, sufficiently risk averse firms establish subsidiaries sooner rather than later. The numerical

illustration shows that this precautionary effect is relatively modest and depends on the composition of cross border frictions, firms' exposure to the destination market, and the cost of changing organisational form.

Several questions remain open. Orbis records the links between parents and subsidiaries, but not whether the parent previously exported to the destination or whether newly established subsidiaries continued to operate after the post-Brexit trading arrangements became clearer. Although we can identify subsidiaries reporting economic activity, we cannot fully assess the extent of their operations or their longer term performance. Nor does the Brexit Uncertainty Index separate news about expected future costs from news about their dispersion. The model establishes how each component affects subsidiary formation and selection, but distinguishing their contributions empirically would require more direct measures of firms' expectations and perceived uncertainty. Beyond Brexit, our findings raise a broader question: whether reorganising corporate structures across borders ahead of a geopolitical shock is a hedge that pays, or a cost firms incur to stand still.

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A. Additional Facts and Results

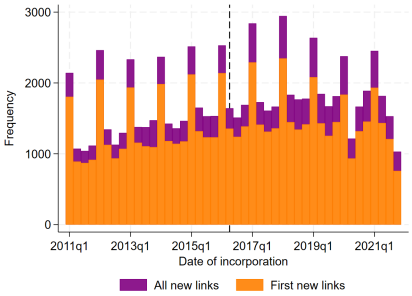
TABLE A.1: Time series of new links frequencies

	UK-EU	**EU-UK**
All new links	18,223	13,269
First new links	14,568	10,857
First ever new links	13,384	10,314
First and only new links	11,473	9,506
First links, no unknown parents	10,003	9,431
First links, no unknown subsidiaries	13,732	10,030
First links, no financial subsidiaries	12,160	9,963
First links, no financial or unknown subsidiaries	11,324	9,136
First links, subsidiaries with balance sheet records	9,601	8,993
First links, parents are also GUO (25%) and GUO (50%)	6,087	4,762

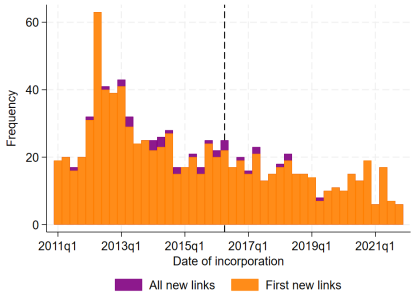
Notes: Period 2011m1-2021m12. Excludes parent firms from sectors 'financial' and 'holdings'. GUO refers to global ultimate owner. Data source *Orbis Historical*.

FIGURE A.1: New links and first new links

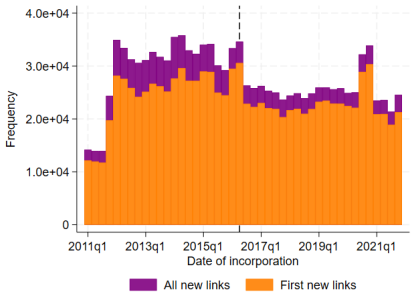
(A) EU → EU



(B) UK → NOCH



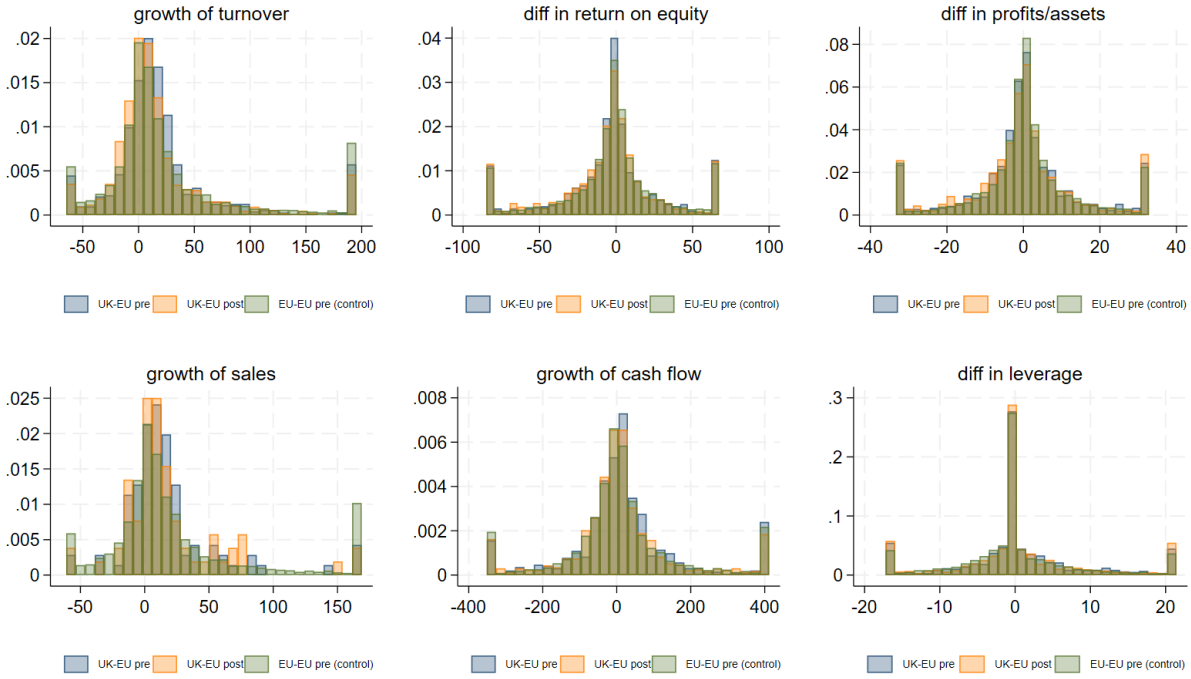
(C) UK → UK



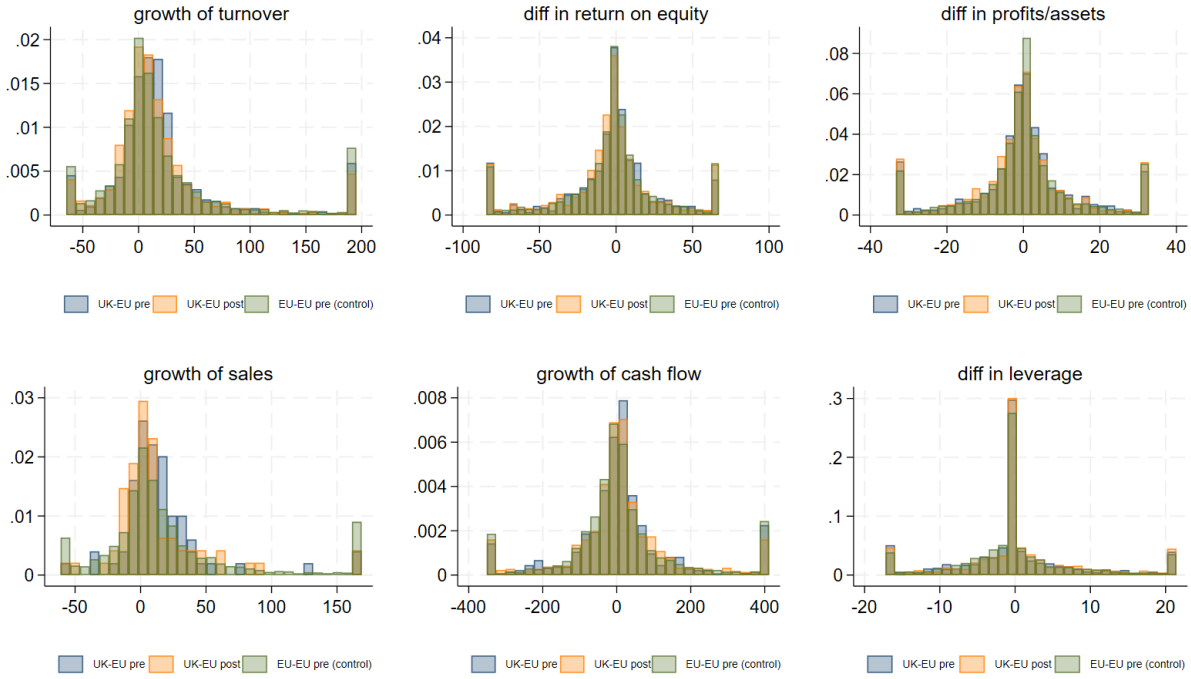
Notes: New parent-subsidiary links created by a EU parent in a EU country (panel (A)), by a UK parent in Norway or Switzerland (panel (B)), by a UK parent in UK (panel (C)). Purple bars denote all new links and orange bars all first new links. The dotted vertical line marks 2016q2, identifying the quarter of the Brexit referendum. Data source *Orbis Historical*, time period 2011q1-2021q4.

FIGURE A.2: Balance sheet distributions of parent firms, sample UK-EU

(A) One year before the first subsidiary incorporation



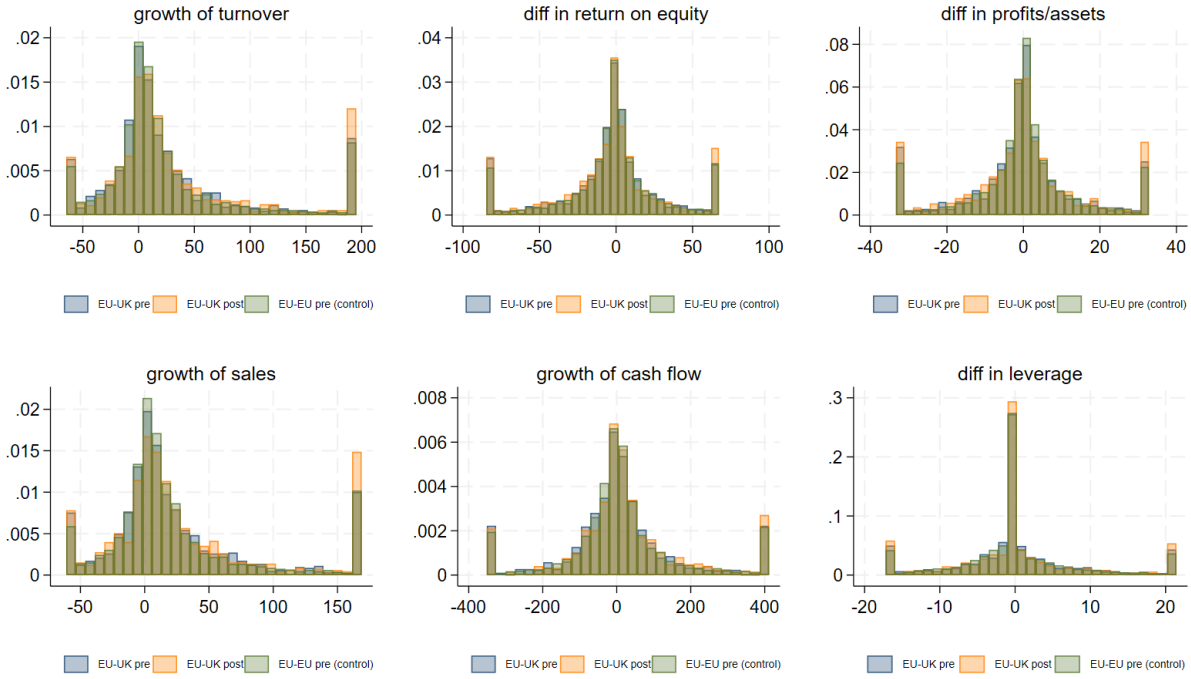
(B) Two years before the first subsidiary incorporation



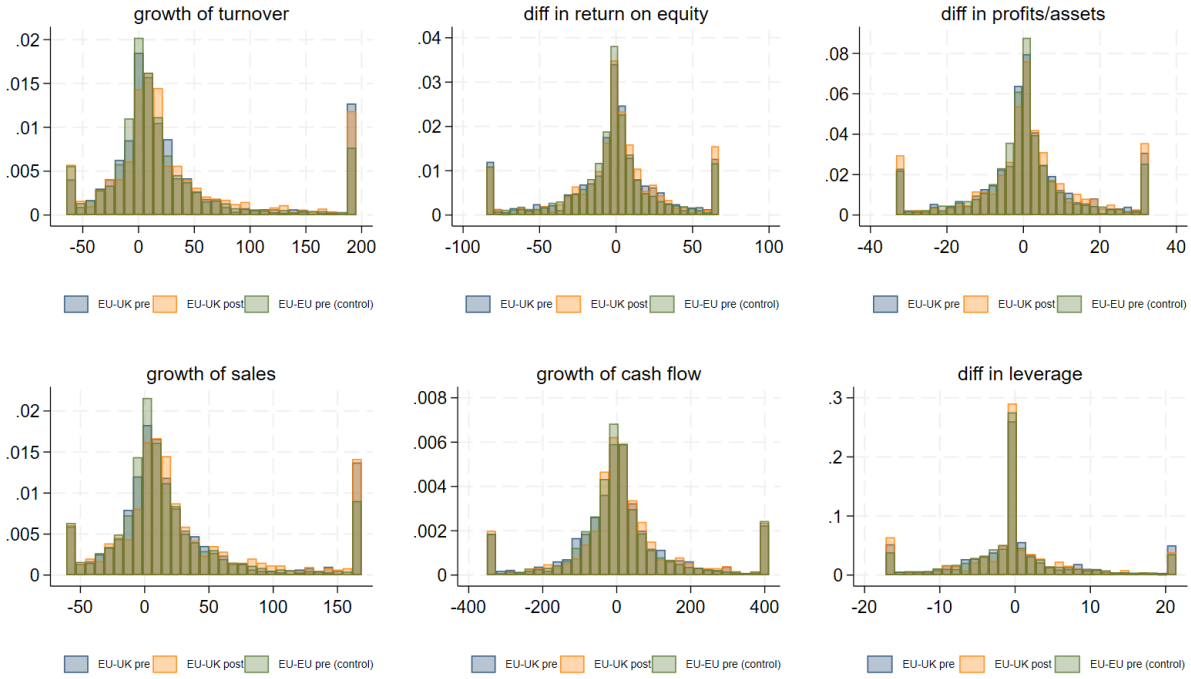
Note: Year-on-year balance sheet variables for parent firms two and one year before they establish a first subsidiary, pre and post Brexit referendum. Variables are winsorised at the 5th and 95th percentiles. Data is from *Orbis Historical*, 2011-2021.

FIGURE A.3: Balance sheet distributions of parent firms, sample EU-UK

(A) One year before the first subsidiary incorporation



(B) Two years before the first subsidiary incorporation



Note: Year-on-year balance sheet variables for parent firms two and one year before they establish a first subsidiary, pre and post Brexit referendum. Variables are winsorised at the 5th and 95th percentiles. Data is from *Orbis Historical*, 2011-2021.

TABLE A.2: Balance sheet characteristics of UK parent firms with first subsidiary, full EU sample

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-11.353***	-11.279***	-12.282***	-6.150***
	(1.699)	(1.553)	(1.547)	(1.313)
Observations	15,204	16,788	12,597	13,834
R^2	0.003	0.004	0.002	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-1.638***	-2.490***	-5.724**	-6.771**
	(0.445)	(0.283)	(2.401)	(2.613)
Observations	16,446	18,053	14,524	15,742
R^2	0.002	0.001	0.001	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.863***	-0.496***	0.741***	0.274*
	(0.167)	(0.123)	(0.128)	(0.142)
Observations	18,304	20,126	15,224	16,682
R^2	0.001	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.3: Balance sheet characteristics of EU parent firms with first subsidiary, full EU sample

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	5.883*	4.399*	7.864**	4.122***
	(3.774)	(2.755)	(2.869)	(1.476)
Observations	15,559	17,257	14,275	15,726
R^2	0.003	0.004	0.003	0.003
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.008**	-1.367	5.338	6.550
	(1.216)	(1.216)	(5.485)	(6.801)
Observations	16,964	18,725	14,985	16,270
R^2	0.003	0.001	0.001	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.075*	-0.598	-0.339*	-0.161
	(0.526)	(0.601)	(0.184)	(0.265)
Observations	18,792	20,764	15,418	16,941
R^2	0.001	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.4: Balance sheet characteristics of UK parent firms with first and only subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-5.122**	-8.529***	-5.301**	-8.976***
	(1.907)	(1.689)	(2.204)	(1.664)
Observations	7,062	7,803	5,400	5,914
R^2	0.002	0.003	0.001	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	0.144	-1.288***	2.102	-3.265
	(0.399)	(0.461)	(2.558)	(3.324)
Observations	7,597	8,320	6,995	7,609
R^2	0.004	0.001	0.001	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.354	-0.030	0.875***	0.420*
	(0.216)	(0.176)	(0.198)	(0.228)
Observations	8,488	9,331	7,340	7,935
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements.* Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.5: Balance sheet characteristics of EU parent firms with first and only subsidiary

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.198***	5.471**	10.980***	3.692**
	(2.964)	(2.300)	(2.252)	(1.439)
Observations	8,242	9,160	6,789	7,478
R^2	0.002	0.002	0.004	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	4.780***	-1.452	12.989**	7.992
	(1.319)	(1.596)	(6.160)	(6.029)
Observations	8,069	8,930	7,400	8,085
R^2	0.006	0.002	0.001	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.590***	-0.603	-0.452	-0.158
	(0.548)	(0.661)	(0.277)	(0.355)
Observations	8,945	9,923	7,483	8,120
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements.* Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.6: Balance sheet characteristics of UK parent firms with first subsidiary, excl. 2016 and 2017

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-8.071***	-5.333***	-11.056***	-1.447
	(1.701)	(1.831)	(1.906)	(1.939)
Observations	7,631	8,456	5,914	6,494
R^2	0.003	0.002	0.002	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-1.204**	-2.421***	-3.736	-2.861
	(0.471)	(0.603)	(3.786)	(3.682)
Observations	8,238	9,042	7,589	8,266
R^2	0.005	0.002	0.001	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.721***	-0.769***	1.191***	0.291
	(0.193)	(0.258)	(0.168)	(0.202)
Observations	9,159	10,077	7,897	8,531
R^2	0.003	0.001	0.002	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.7: Balance sheet characteristics of EU parent firms with first subsidiary, excl. 2016 and 2017

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.048***	11.001***	11.800***	7.461***
	(2.429)	(2.720)	(2.340)	(1.391)
Observations	7,947	8,870	7,289	8,051
R^2	0.004	0.003	0.005	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.681***	-1.114	26.037***	10.108
	(0.891)	(1.295)	(5.627)	(9.276)
Observations	8,675	9,606	7,987	8,725
R^2	0.006	0.001	0.003	0.001
	$\Delta\left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta\left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	0.940	-0.356	-0.042	-0.349
	(0.592)	(0.691)	(0.257)	(0.362)
Observations	9,575	10,620	8,107	8,804
R^2	0.003	0.001	0.001	0.001

Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.8: Balance sheet characteristics of UK parent firms with first subsidiary, excl. parents with unknown sector

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times UKEU$	-4.563***	-8.154***	-7.303***	-4.003**
	(1.593)	(1.667)	(1.828)	(1.776)
Observations	8,685	9,592	6,603	7,220
R^2	0.002	0.003	0.002	0.001
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times UKEU$	-0.692**	-2.210***	2.013	-3.958
	(0.332)	(0.437)	(2.833)	(3.716)
Observations	9,363	10,249	8,596	9,356
R^2	0.004	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta \left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times UKEU$	-0.562***	-0.235	0.917***	0.284
	(0.197)	(0.172)	(0.165)	(0.215)
Observations	10,412	11,437	8,978	9,699
R^2	0.002	0.001	0.001	0.001

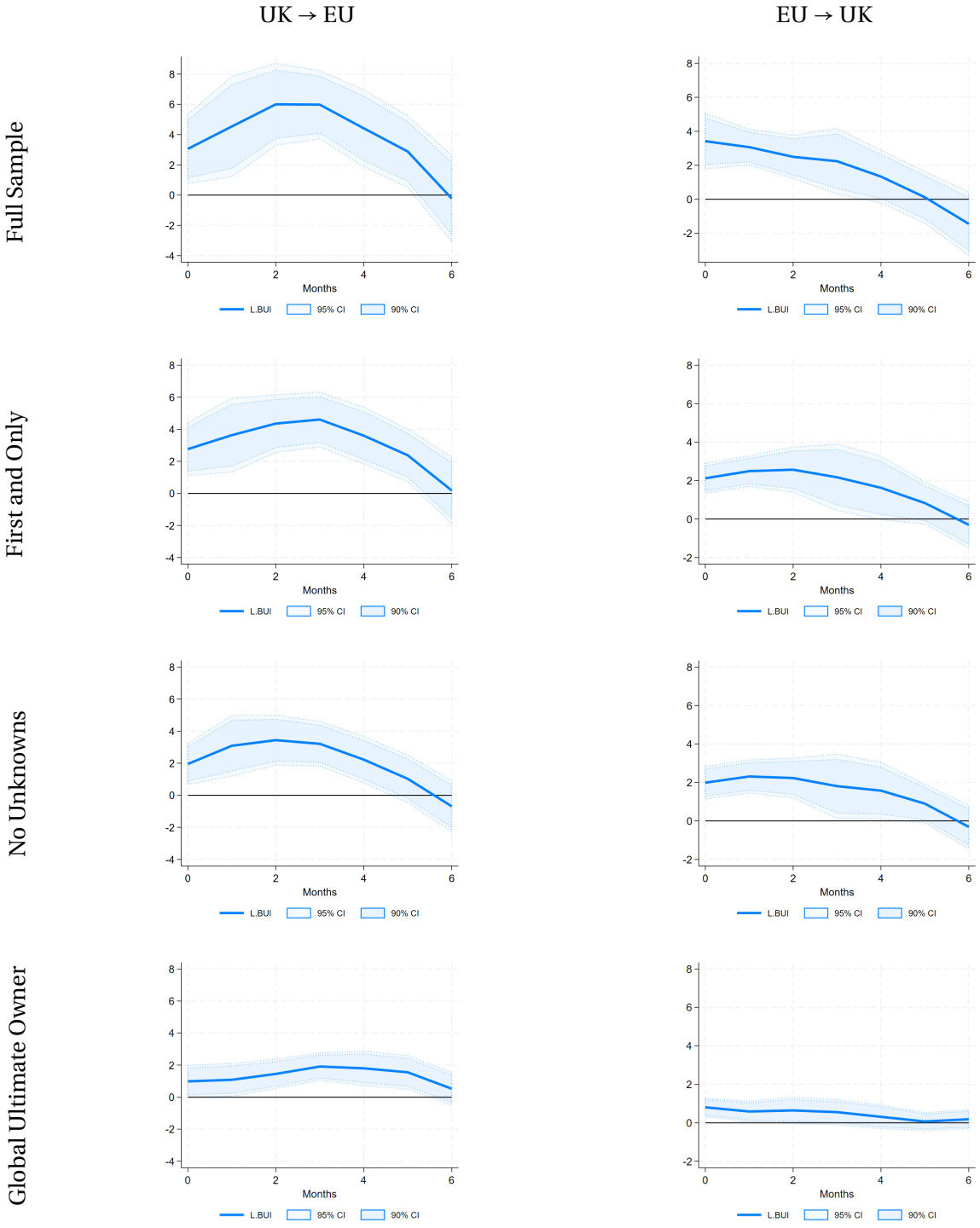
Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (1)-(2). Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A.9: Balance sheet characteristics of EU parent firms with first subsidiary, excl. parents with unknown sector

	$\Delta \log TOver_{-2}$	$\Delta \log TOver_{-1}$	$\Delta \log Sales_{-2}$	$\Delta \log Sales_{-1}$
$POST \times EUUK$	11.394***	8.115***	11.958***	6.508***
	(2.500)	(2.287)	(2.117)	(1.364)
Observations	9,035	10,048	8,274	9,102
R^2	0.004	0.002	0.006	0.002
	ΔROE_{-2}	ΔROE_{-1}	$\Delta \log CFlow_{-2}$	$\Delta \log CFlow_{-1}$
$POST \times EUUK$	3.876***	-1.179	11.723**	11.535
	(1.352)	(1.375)	(4.941)	(7.295)
Observations	9,877	10,911	9,049	9,877
R^2	0.005	0.002	0.001	0.001
	$\Delta \left(\frac{Profits}{Assets}\right)_{-2}$	$\Delta \left(\frac{Profits}{Assets}\right)_{-1}$	$\Delta Leverage_{-2}$	$\Delta Leverage_{-1}$
$POST \times EUUK$	1.506***	-0.260	-0.209	-0.104
	(0.514)	(0.621)	(0.182)	(0.328)
Observations	10,889	12,056	9,157	9,943
R^2	0.002	0.001	0.001	0.001

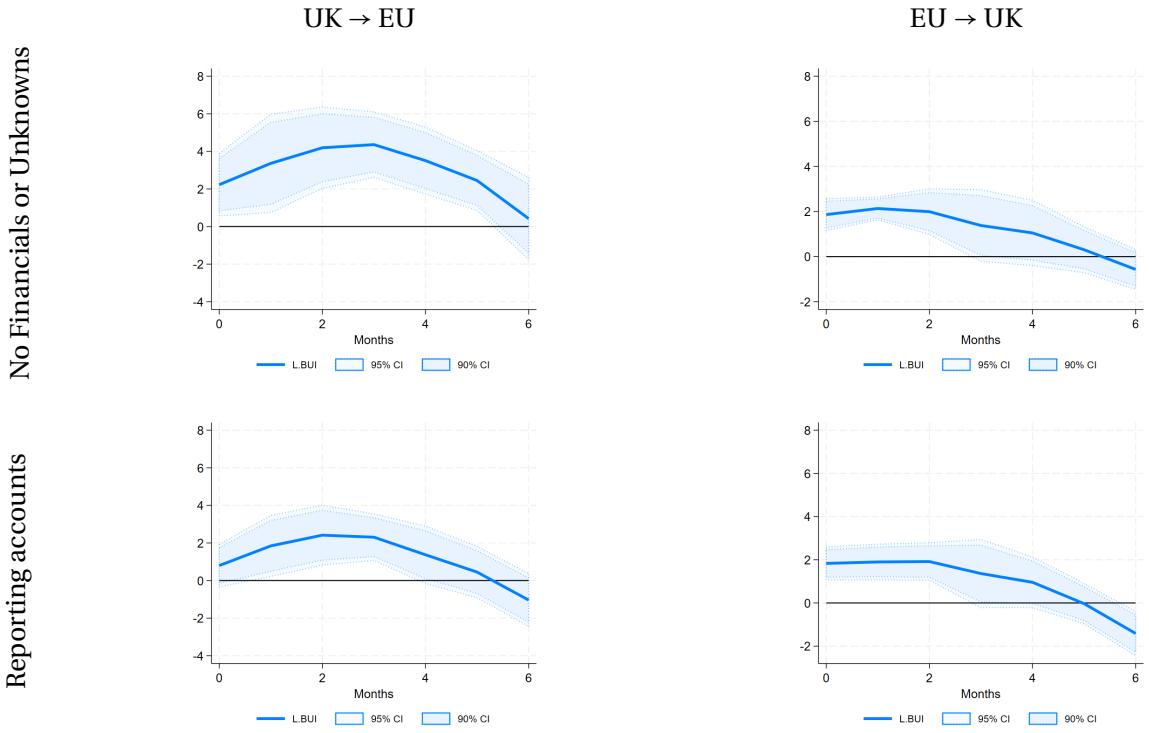
Notes: Estimated coefficients β_{-2} and β_{-1} from specifications (3)-(4).Sample 2011-2021, yearly frequency. Data sources *Orbis Historical, Office of National Statistics UK, Eurostat* and *Bank of International Settlements*. Standard errors in parentheses and clustered at parent country level. Statistical significance * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

FIGURE A.4: Dynamic responses of new links to uncertainty: Robustness I - Parent charcteristics



Note: LPs of new links on (i) BUI_{t-1}, (ii) CUI_{t-1}, (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical*, *Office of National Statistics UK*, and *Eurostat*.

FIGURE A.5: Dynamic responses of new links to uncertainty: Robustness II - Subsidiary characteristics



Note: LPs of new links on (i) BUI_{t-1}, (ii) CUI_{t-1}, (iii) macro controls (growth rate of real GDP of UK or EU, growth rate of real exports from UK to EU, or EU to UK, growth rate of real effective exchange rate of UK or EU) (v) year dummy variables, (vi) Jan and Aug month dummies. Time period 2016m9-2021m12, CIs at 95% and 90% significance level, based on robust standard errors. Data sources *Orbis Historical, Office of National Statistics UK*, and *Eurostat*.

TABLE A.10: Sectors of parent firms, UK-EU and EU-UK samples

	Parent sector UK-EU sample			**Parent sector EU-UK sample**		
	Pre-Brexit 2011-2016	*Post-Brexit* 2016-2021	*Total* 2011-2021	*Pre-Brexit* 2011-2016	*Post-Brexit* 2016-2021	*Total* 2011-2021
Accomodation	100	65	165	62	50	112
Administrative	1,459	1,151	2,610	538	552	1,090
Agriculture	19	18	37	27	34	61
Construction	159	215	374	253	204	457
Education	34	44	78	32	34	66
Electricity	65	82	147	216	67	283
Extraterritorial	33	15	48	1	0	1
Health	53	55	108	31	61	92
Household	4	6	10	0	0	0
Information	796	960	1,756	896	807	1,703
Manufacturing	559	592	1,151	818	655	1,473
Mining	90	47	137	28	20	48
Other	297	261	558	236	462	698
Professional	1,441	1,628	3,069	1,659	1,350	3,009
Public	7	0	7	2	3	5
Real estate	330	312	642	236	213	449
Recreation	65	61	126	50	58	108
Retail	616	733	1,349	889	796	1,685
Transportation	164	157	321	155	139	294
Water	17	6	23	18	20	38
Unknown	2,056	3,451	5,507	758	839	1,597
Total	**8,364**	**9,859**	**18,223**	**6,905**	**6,364**	**13,269**

Notes: Number of links in UK-EU and EU-UK by parent sector, pre and post Brexit referendum. Sectors based on NACE rev. 2 classification, and authors' own sectoral refinements, based on trade description text and company registered names. Excludes financial sector and holding companies of unknown classification. Data source *Orbis Historical*.

B. Orbis Financials Dataset

We utilize firm-level data from Bureau van Dijk's Orbis Historical, merging data on firmographics (including firm characteristics like name, age and sector) with balance sheet financials. The following sets out the precise cleaning procedure, which broadly follows Kalemli-Ozcan et al. (2024).

For the balance sheet variables, we aim for unique firm-financial year observations. Therefore, we first establish unique observations in terms of **BvD ID, financial year, consolidation code and filing type** by keeping the latest observation, i.e. a reporting month closer to December, and a reporting day closer to 31. Second, to get unique observations in terms of **BvD ID, year, and consolidation code**, we drop duplicate rows of these three variables where the filing type is local registry filings in favor of annual reports. This is because local registry filings are more often coded as limited financials. Third, to get unique **BvD ID and year** observations, we deal with consolidation codes. Following Kalemli-Ozcan et al. (2024), we keep unconsolidated over consolidated accounts to avoid double counting. Within consolidated accounts, we keep those where the company headquarter does not present an unconsolidated account. Within unconsolidated accounts, we keep those where the company headquarter does not present a consolidated account. Next, we treat same-year reporting by keeping the observation with the reporting month closer to December. Similar, for same-month reporting, we keep the reporting day closer to the end of the month. Finally, for BvD ID-year observations which differ only in terms of original reporting units, we keep the observations in "units" as this is the most frequent reporting unit.

Having eliminated duplicates, we eliminate likely reporting errors, employing basic data cleaning following Kalemli-Ozcan et al. (2024). As such, we drop firms with negative total assets, sales, fixed assets, tangible fixed assets, number of employees, current or non-current liabilities. Furthermore, we drop observations with more than 2 million employees. We also drop observations where total assets *and* operating revenue *and* sales *and* employment are jointly missing. Observations with a negative leverage ratio, computed as (long-term debt + loans)/ total assets, are dropped. Finally, we construct an indicator for inconsistent balance sheets by comparing total assets to the sum of shareholder funds, current and non-current liabilities. This ratio should equal 1 for consistency, and we drop observations with a ratio below the 1st or above the 99th percentiles.

C. The model: definitions, proofs and numerical method

This appendix follows the order of Section 6. Appendix C.1 sets out the switching problem, the valuation of risky profit and the conditions under which the decision is described by a single cutoff. Appendix C.2 derives the four-term decomposition (9) and defines the inaction zone. Appendix C.3 characterises the cutoff under known costs and proves condition (10). Appendix C.4 proves the results on uncertainty, including conditions (11) and (13). Appendix C.5 studies the option term outside the inaction zone. Appendix C.6 documents the calibration, the numerical method and the experiments of Section 6.2.3.

Throughout, $x^+ \equiv \max\{x,0\}$ and $\mathbb{E}_0$ is the expectation given information at $t=0$. Let $\theta(\tau) \equiv \tau^{1-\sigma}$ and note that since $\sigma > 1$, $\theta' < 0$ and $\theta'' > 0$. For a firm of capability ϕ let

$$Z(\phi;\tau,F_I) \equiv \Pi_I(\phi;F_I) - \Pi_X(\phi;\tau) = \phi\left[\alpha - \theta(\tau)\right] - F_I \tag{C.1}$$

be the advantage of local presence at costs (τ, F_I), with Π_X and Π_I as in (7). Three values of Z recur: the current advantage $Z_0(\phi) \equiv Z(\phi;\tau_0,F_{I,0})$, the advantage at expected future costs $\bar{Z}_1(\phi) \equiv Z(\phi;\mu_\tau,\mu_F)$, and the realised advantage $\tilde{Z}_1(\phi) \equiv Z(\phi;\tau_1,F_{I,1})$, which is random as of $t=0$. We write $\tilde{\Pi}_X \equiv \Pi_X(\phi;\tau_1)$ and $\tilde{\Pi}_I \equiv \Pi_I(\phi;F_{I,1})$ for realised period-1 operating profits before any switch, so that $\tilde{Z}_1 = \tilde{\Pi}_I - \tilde{\Pi}_X$. In the fixed-friction extension of Section 6.2.2 the cross-border mode also bears a per-period fixed cost $F_{X,1} = \mu_X + k\varepsilon_\tau$ in period 1, with $F_{X,0} = 0$ in period 0 and $\mu_X = k = 0$ in the benchmark. Period-1 cross-border profit is then $\pi_D + \phi\theta(\tau_1) - F_{X,1}$, so that $\bar{Z}_1$ includes $+\mu_X$ and $\tilde{Z}_1$ includes $+F_{X,1}$. Every definition and result below applies with Π_X read in this way, and the values of μ_X and k are given in Appendix C.6.

C.1 Timing, valuation and the cutoff (Section 6.1)

Sorting condition. We impose

$$\alpha > \theta(\tau) \quad \text{for all relevant } \tau, \tag{C.2}$$

so that the variable-profit advantage of a subsidiary, $\phi[\alpha - \theta(\tau)]$, is positive and increasing in capability. This is the condition that sorts firms by capability in Helpman et al. (2004).

The switching problem. At $t=1$ the realised costs $(\tau_1, F_{I,1})$ are observed and a firm in either mode can change mode by paying κ. The period 1 profits of a firm that starts $t=1$ in mode X (a *waiter*) or in mode I (a *parent*) are therefore

$$\Pi_{X,1}(\phi) = \max\left\{\tilde{\Pi}_X, \tilde{\Pi}_I - \kappa\right\}, \qquad \Pi_{I,1}(\phi) = \max\left\{\tilde{\Pi}_I, \tilde{\Pi}_X - \kappa\right\}. \tag{C.3}$$

The firm always serves the destination: serving it from abroad earns a positive operating profit and carries no fixed cost, so withdrawing is never optimal, and in each period the firm is either in X or in I. Using $\max\{a, b-\kappa\} = a + (b-a-\kappa)^+$ state by state,

$$\Pi_{X,1} = \tilde{\Pi}_X + (\tilde{Z}_1 - \kappa)^+, \qquad \Pi_{I,1} = \tilde{\Pi}_I + (-\tilde{Z}_1 - \kappa)^+. \tag{C.4}$$

A waiter switches to I only if $\tilde{Z}_1 > \kappa$, gaining $\tilde{Z}_1 - \kappa$ and a parent switches back only if $\tilde{Z}_1 < -\kappa$, gaining $-\tilde{Z}_1 - \kappa$.

Valuation. Period 1 profit is random, and the firm values it by its certainty equivalent,

$$CE_0(Y) \equiv u^{-1}\big(\mathbb{E}_0[u(Y)]\big), \qquad u(\Pi) = \begin{cases} \frac{Y^{1-\gamma}}{1-\gamma} & \text{if } \gamma \geq 0 \text{ and } \gamma \neq 1 \\ ln(Y) & \text{if } \gamma = 1 \end{cases}. \tag{C.5}$$

When $\gamma = 0$, $CE_0 = \mathbb{E}_0$. The value at $t = 0$ of choosing mode $M \in \{X, I\}$ is $V_{M,0} = \Pi_{M,0} + \delta\, CE_0(\Pi_{M,1})$ with $\delta \in (0,1]$, $\Pi_{X,0} = \Pi_X(\phi;\tau_0)$ and $\Pi_{I,0} = \Pi_I(\phi;F_{I,0}) - \kappa$, and the gain from establishing the subsidiary now is $D_0(\phi) \equiv V_{I,0}(\phi) - V_{X,0}(\phi)$, which is (8). The parameter γ can be more general than the risk aversion of the firm's owners: financing frictions and distress costs, which make low-profit states especially costly, have the same effect.

The certainty equivalent is used for three reasons. It ranks risky payoffs exactly as expected utility does; it is measured in units of profit, like current profit and the switching cost, so the three can be added; and the certainty equivalent of a sure payoff is the payoff itself, so that when costs are known the firm maximises the present value of profit whatever the value of γ, and κ is valued at its monetary cost. The third property gives a benchmark that does not depend on γ, to which the effects of uncertainty are added one at a time in (9). Two natural alternatives lack it. Adding expected utility to current profit, $\Pi_0 + \delta\mathbb{E}_0[u(\Pi_1)]$, adds utility to profit: a unit of future profit is then weighted by $\delta u'(\Pi_1)$ rather than δ, so γ moves the cutoff even when nothing is random and the cutoff depends on the units in which profit is measured. Time-separable expected utility, $u(\Pi_0) + \delta\mathbb{E}_0[u(\Pi_1)]$, has consistent units, but one parameter then governs both the aversion to risk and the willingness to move profit between periods, so γ again matters without risk. Solving the model under the second alternative changes no sign in Table 5 and moves its entries by a few hundredths of a percentage point. The decision is described by a cutoff when D_0 is increasing in ϕ.

Lemma C.1 (Sorting)**.** *Suppose* (C.2) *holds and either future costs are known or* $\gamma = 0$*. Then* D_0 *is strictly increasing in* ϕ*.*

Proof. In both cases $CE_0 = \mathbb{E}_0$, so by (C.4) $D_0(\phi) = Z_0(\phi) - \kappa + \delta\,\mathbb{E}_0[g(\tilde{Z}_1(\phi))]$ with g as in (C.6) below. The first term is strictly increasing in ϕ by (C.2). In every state $\tilde{Z}_1(\phi)$ is strictly increasing in ϕ, with slope $\alpha - \theta(\tau_1) > 0$, and g is non-decreasing, so $\mathbb{E}_0[g(\tilde{Z}_1(\phi))]$ is non-decreasing. Away from the kinks of g,

$$D_{0,\phi} = [\alpha - \theta(\tau_0)] + \delta\,\mathbb{E}_0\big[(\alpha - \theta(\tau_1))\,\mathbf{1}\{|\tilde{Z}_1| < \kappa\}\big] > 0. \qquad \square$$

Under risk aversion, $d\,CE_0(Y(\phi))/d\phi = \mathbb{E}_0[u'(Y)\,Y_\phi]/u'(CE_0(Y))$, and the marginal-utility weights differ across modes, so Lemma C.1 does not carry over directly. Two results hold. First, under local inaction (Definition C.1) D_0 and $D_{0,\phi}$ are continuous in $(\sigma_\tau^2, \sigma_F^2)$ and coincide with their known-cost counterparts at zero, so $D_{0,\phi} > 0$ near the cutoff for small risk. Second, if $\pi_D \equiv 0$, $\varepsilon_F = 0$ and local inaction holds, the homogeneity of CRRA utility gives $CE_0(\phi\theta(\tau_1)) = \phi\, CE_0(\theta(\tau_1))$, so $D_{0,\phi} = [\alpha - \theta(\tau_0)] + \delta[\alpha - CE_0(\theta(\tau_1))] > 0$ whenever $\alpha > \sup\theta(\tau_1)$. For larger risks, general π_D or switching, we assume that D_0

is increasing in ϕ at the marginal firm; in every numerical experiment of Section 6.2.3 the root of D_0 is unique and D_0 changes sign once on the bracket searched.

Subsidiary formation and selection. Capability is distributed according to G, and the mass of new parents in a cohort is $M_I = 1 - G(\phi_I^*)$.

Lemma C.2 (Subsidiary formation and selection)**.** *A lower cutoff raises the mass of new parents and lowers the mean, among them, of any characteristic $h(\phi)$ increasing in capability; a higher cutoff does the opposite.*

Proof. $M_I = 1 - G(\phi_I^*)$ is non-increasing in ϕ_I^*. For h increasing and G differentiable, $\frac{d}{dc}E[h(\phi) \mid \phi \geq c] = \frac{G'(c)}{1-G(c)}\big(E[h(\phi) \mid \phi \geq c] - h(c)\big) \geq 0$. □

The number and the composition of new parents are therefore governed by the same cutoff and move in opposite directions, whatever moves the cutoff: every result below about the subsidiary-formation cutoff ϕ_I^* is also a result about the number and the selection of new parents. The pre-incorporation measures of Section 3 are read as characteristics increasing in capability; the lemma requires only that subsidiary formation be ordered by a single index.

C.2 The decomposition and the inaction zone (Section 6.2)

Starting period 1 with or without a subsidiary. Compare, in period 1, a parent and a waiter of the same capability. By (C.4), $\Pi_{I,1} - \Pi_{X,1} = \tilde{Z}_1 - (\tilde{Z}_1 - \kappa)^+ + (-\tilde{Z}_1 - \kappa)^+ = g(\tilde{Z}_1)$, where

$$g(z) \equiv \begin{cases} -\kappa & \text{if } z \leq -\kappa \quad \text{(the parent switches back to } X\text{)}, \\ z & \text{if } -\kappa < z < \kappa \quad \text{(no firm switches)}, \\ \kappa & \text{if } z \geq \kappa \quad \text{(the waiter switches to } I\text{)}. \end{cases} \tag{C.6}$$

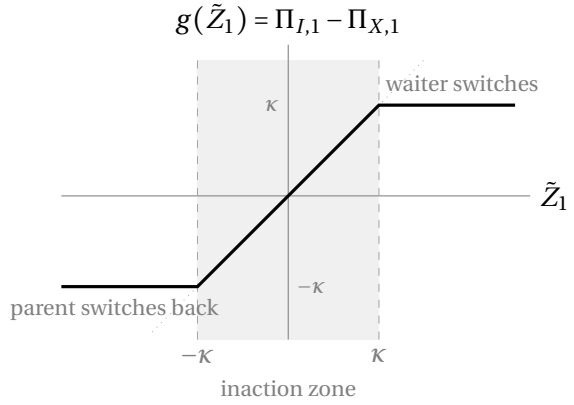
Starting with a subsidiary is worth the realised advantage of local presence capped at the switching cost in either direction (Figure C.1): when $\tilde{Z}_1 > \kappa$ the waiter switches and the head start is worth only the switching cost saved; when $\tilde{Z}_1 < -\kappa$ the parent closes and the subsidiary is a liability equal to the cost of closing it; in between neither switches and the difference is $\tilde{Z}_1$ itself.

Option values and risk premia. Define

$$\Omega_X \equiv \mathbb{E}_0\big[(\tilde{Z}_1 - \kappa)^+\big], \qquad \Omega_I \equiv \mathbb{E}_0\big[(-\tilde{Z}_1 - \kappa)^+\big], \qquad RP(Y) \equiv \mathbb{E}_0[Y] - CE_0(Y). \tag{C.7}$$

Ω_X is the value at $t = 0$, to a firm that waits, of being able to establish the subsidiary at $t = 1$ once costs are known; Ω_I is the value, to a firm that enters now, of being able to close the subsidiary. $RP(Y) \geq 0$ is the risk premium of the payoff Y: it is zero if Y is not random or if $\gamma = 0$, and positive otherwise by the concavity of u.

FIGURE C.1: Starting period 1 with a subsidiary (in mode I)



Notes: In the inaction zone (shaded area) neither firm switches and the difference equals the realised advantage of local presence, $\tilde{Z}_1$. Outside it one of the two firms switches and the difference is capped at $\pm\kappa$.

Derivation of (9). Taking expectations of (C.4),

$$\mathbb{E}_0\left[\Pi_{I,1} - \Pi_{X,1}\right] = \mathbb{E}_0\left[\tilde{Z}_1\right] - (\Omega_X - \Omega_I), \qquad \mathbb{E}_0\left[\tilde{Z}_1\right] = \phi\left[\alpha - \mathbb{E}_0\theta(\tau_1)\right] - \mu_F = \bar{Z}_1(\phi) - J(\phi),$$

with $J(\phi) \equiv \phi[\mathbb{E}_0\theta(\tau_1) - \theta(\mu_\tau)]$. By definition $CE_0(Y) = \mathbb{E}_0[Y] - RP(Y)$, hence

$$CE_0(\Pi_{I,1}) - CE_0(\Pi_{X,1}) = \bar{Z}_1 - J + \underbrace{\left[RP(\Pi_{X,1}) - RP(\Pi_{I,1})\right]}_{P} - \underbrace{(\Omega_X - \Omega_I)}_{W}, \tag{C.8}$$

which, substituted in (8), is (9). The identity is exact and holds whether or not firms switch in period 1.

We next describe properties of the terms in the decomposition. $J \geq 0$ by Jensen's inequality, since θ is convex, and J weakly rises under a mean-preserving spread of τ_1. The convexity arises because the firm sets its price after observing τ; a per-period fixed cost of serving from abroad enters profit linearly and contributes nothing to J (Appendix C.4). Subsidiary profit is linear in F_I, so uncertainty about $F_{I,1}$ contributes nothing to J either. P is positive when starting period 1 in X is the riskier position and negative when the subsidiary is riskier; it is zero if $\gamma = 0$. If $\kappa = 0$, $\Pi_{X,1} = \Pi_{I,1} = \max\{\tilde{\Pi}_X, \tilde{\Pi}_I\}$ in every state and the left-hand side of (C.8) is zero. If no firm switches in any state, $\Omega_X = \Omega_I = 0$ and $W = 0$. Since $(x-\kappa)^+$ is convex, $\Omega_X \geq (\hat{m}-\kappa)^+$ and $\Omega_I \geq (-\hat{m}-\kappa)^+$ with $\hat{m} \equiv \mathbb{E}_0[\tilde{Z}_1]$: W contains the gain from switching that a firm would plan even if future costs were known, $(\hat{m}-\kappa)^+ - (-\hat{m}-\kappa)^+$, which is zero whenever $|\hat{m}| \leq \kappa$, and only the excess of W over it is the value of waiting for information. With known costs the inequalities hold with equality and $\bar{Z}_1 - W = g(\bar{Z}_1)$.

The inaction zone. We first define the inaction zone:

Definition C.1 (Inaction zone)**.** A firm is in the inaction zone in a given state if $-\kappa < \tilde{Z}_1(\phi) < \kappa$. *Local inaction* holds if the marginal firm ϕ_I^* is in the inaction zone for all relevant realisations of future costs.

Local inaction restricts switching at $t = 1$, not the choice of mode at $t = 0$: whatever mode the marginal firm chooses at $t = 0$, it keeps at $t = 1$. Under local inaction $\Omega_X = \Omega_I = 0$, so $W = 0$, the period-1 payoffs

are $\Pi_{X,1} = \tilde{\Pi}_X$ and $\Pi_{I,1} = \tilde{\Pi}_I$, and each is exposed to one source of risk only. With unbounded shocks local inaction cannot hold in every state. The small-risk results below are unaffected provided $|\bar{Z}_1(\phi_I^*)| < \kappa$, because the probability of switching is then exponentially small in $1/\sigma^2$ and does not enter the second-order terms. The marginal firm is in the zone before the referendum, and Proposition C.1 gives the exact condition under which it stays there after a change in expected costs.

C.3 Expected costs (Section 6.2.1) Suppose future costs are known: $\varepsilon_\tau = \varepsilon_F = 0$, so $J = P = 0$, $\tilde{Z}_1 = \bar{Z}_1$ and, by (C.4), $D_0(\phi) = Z_0(\phi) - \kappa + \delta\, g(\bar{Z}_1(\phi))$, strictly increasing in ϕ by Lemma C.1. In the inaction zone $g(\bar{Z}_1) = \bar{Z}_1$ and

$$D_0(\phi) = Z_0(\phi) - \kappa + \delta\, \bar{Z}_1(\phi): \tag{C.9}$$

a firm that enters now earns the advantage of local presence in both periods and pays κ once. Let

$$M(\phi) \equiv \bar{Z}_1(\phi) - Z_0(\phi) = \phi\big[\theta(\tau_0) - \theta(\mu_\tau)\big] - (\mu_F - F_{I,0}), \qquad H(\mu_\tau) \equiv [\alpha - \theta(\tau_0)] + \delta[\alpha - \theta(\mu_\tau)] > 0 \tag{C.10}$$

be the expected change in the advantage of local presence between the two periods and the slope of (C.9) in ϕ.

Proposition C.1 (Expected costs)**.** *Suppose future costs are known, and let*

$$\phi^z \equiv \frac{F_{I,0} + \kappa + \delta\mu_F}{H(\mu_\tau)}, \qquad \phi^+ \equiv \frac{F_{I,0} + (1-\delta)\kappa}{\alpha - \theta(\tau_0)}, \qquad \phi^- \equiv \frac{F_{I,0} + (1+\delta)\kappa}{\alpha - \theta(\tau_0)}. \tag{C.11}$$

The cutoff is

$$\phi_I^* = \begin{cases} \phi^- & \text{if } M(\phi^z) \le -(2+\delta)\kappa \quad \textit{(a new parent would close the subsidiary at } t = 1\textit{)}, \\ \phi^z & \text{if } -(2+\delta)\kappa < M(\phi^z) < \delta\kappa \quad \textit{(the marginal firm is in the inaction zone)}, \\ \phi^+ & \text{if } M(\phi^z) \ge \delta\kappa \quad \textit{(a waiter would establish the subsidiary at } t = 1\textit{)}. \end{cases} \tag{C.12}$$

In the inaction zone the cutoff is strictly decreasing in μ_τ and strictly increasing in μ_F. Outside it, ϕ^+ and ϕ^- do not depend on μ_τ or μ_F. Hence ϕ_I^* is non-increasing in μ_τ and non-decreasing in μ_F for changes in expected costs of any size.

Proof. The three cases. ϕ^z is the unique root of the linear equation $Z_0(\phi) - \kappa + \delta\bar{Z}_1(\phi) = 0$; rearranging gives the first expression in (C.11). Since $Z_0 = \bar{Z}_1 - M$, the same equation gives $\bar{Z}_1(\phi^z) = (\kappa + M(\phi^z))/(1+\delta)$, so

$$\bar{Z}_1(\phi^z) < \kappa \iff M(\phi^z) < \delta\kappa, \qquad \bar{Z}_1(\phi^z) > -\kappa \iff M(\phi^z) > -(2+\delta)\kappa.$$

Middle case. If $|\bar{Z}_1(\phi^z)| < \kappa$ then $g(\bar{Z}_1(\phi^z)) = \bar{Z}_1(\phi^z)$, so $D_0(\phi^z) = 0$ and $\phi_I^* = \phi^z$. Conversely, if $|\bar{Z}_1(\phi_I^*)| < \kappa$ then ϕ_I^* solves the linear equation and equals ϕ^z. The marginal firm is therefore in the inaction zone if and only if $-(2+\delta)\kappa < M(\phi^z) < \delta\kappa$. *Third case.* If $M(\phi^z) \ge \delta\kappa$ then $\bar{Z}_1(\phi^z) \ge \kappa$ and $g(\bar{Z}_1(\phi^z)) =$

$\kappa \le \bar{Z}_1(\phi^z)$, so $D_0(\phi^z) \le 0$ and $\phi_I^* \ge \phi^z$. Since $\bar{Z}_1$ is increasing in ϕ, $\bar{Z}_1(\phi) \ge \kappa$ for every $\phi \ge \phi^z$, where $D_0(\phi) = Z_0(\phi) - (1-\delta)\kappa$, whose root is ϕ^+: a waiter establishes the subsidiary at $t=1$, so the firm has a subsidiary in period 1 whatever it does at $t=0$, and entering now only earns the current advantage and brings forward the payment of κ. *First case.* If $M(\phi^z) \le -(2+\delta)\kappa$ then $\bar{Z}_1(\phi^z) \le -\kappa$ and $g(\bar{Z}_1(\phi^z)) = -\kappa \ge \bar{Z}_1(\phi^z)$, so $D_0(\phi^z) \ge 0$ and $\phi_I^* \le \phi^z$. For every $\phi \le \phi^z$, $\bar{Z}_1(\phi) \le -\kappa$ and $D_0(\phi) = Z_0(\phi) - (1+\delta)\kappa$, whose root is ϕ^-: a new parent closes the subsidiary at $t=1$, earning the current advantage for one period and paying κ twice. The cutoff is continuous where the cases meet: at $M(\phi^z) = \delta\kappa$, $Z_0(\phi^z) = \bar{Z}_1 - M = (1-\delta)\kappa$, so $\phi^z = \phi^+$; at $M(\phi^z) = -(2+\delta)\kappa$, $Z_0(\phi^z) = (1+\delta)\kappa$, so $\phi^z = \phi^-$.

Response to expected costs. In the zone, differentiating (C.11),

$$\frac{\partial \phi_I^*}{\partial \mu_F} = \frac{\delta}{H} > 0, \qquad \frac{\partial \phi_I^*}{\partial \mu_\tau} = \frac{(F_{I,0} + \kappa + \delta\mu_F)\,\delta\,\theta'(\mu_\tau)}{H^2} < 0. \tag{C.13}$$

Outside the zone the cutoff is ϕ^+ or ϕ^-, and neither contains μ_τ or μ_F. For changes of any size, let $\mu_\tau' > \mu_\tau$. Then $\theta(\mu_\tau') < \theta(\mu_\tau)$ and $\bar{Z}_1'(\phi) > \bar{Z}_1(\phi)$ for every ϕ; since g is non-decreasing, $D_0'(\phi) \ge D_0(\phi)$ for every ϕ, so $D_0'(\phi_I^*) \ge 0$ at the old cutoff and by Lemma C.1 the new cutoff is weakly lower, whichever case applies before and after. The argument for μ_F is symmetric. □

Condition (10). For a change from (μ_τ^0, μ_F^0) to (μ_τ^1, μ_F^1) that keeps the marginal firm in the zone, (C.9) evaluated at the old cutoff gives

$$D_0^1(\phi_I^{*,0}) - D_0^0(\phi_I^{*,0}) = \delta\Big\{\phi_I^{*,0}\big[\theta(\mu_\tau^0) - \theta(\mu_\tau^1)\big] - (\mu_F^1 - \mu_F^0)\Big\},$$

and since D_0 is increasing in ϕ the cutoff falls if and only if the term in braces is positive. Starting from the status quo, $(\mu_\tau^0, \mu_F^0) = (\tau_0, F_{I,0})$, the term in braces is $M(\phi_I^{*,0})$, which is (10). Because M is increasing in ϕ when $\theta(\tau_0) > \theta(\mu_\tau)$, the condition can equivalently be checked at the new cutoff, as in (C.12).

C.4 Uncertainty (Section 6.2.2)

Throughout this subsection local inaction holds, so $W = 0$ and (9) reads

$$D_0(\phi) = Z_0(\phi) - \kappa + \delta\big[\bar{Z}_1(\phi) - J(\phi) + P(\phi)\big], \tag{C.14}$$

with $\Pi_{X,1} = \pi_D(\phi) + \phi\theta(\tau_1)$ exposed only to the trade cost and $\Pi_{I,1} = \pi_D(\phi) + \alpha\phi - F_{I,1}$ only to the subsidiary cost. The shocks have finite third moments. Unless stated otherwise θ, θ' and θ'' are evaluated at μ_τ; with CES profits $\theta'(\tau) = -(\sigma-1)\theta(\tau)/\tau < 0$ and $\theta''(\tau) = \sigma(\sigma-1)\theta(\tau)/\tau^2 > 0$.

The convexity term. Expanding θ around μ_τ and taking expectations,

$$\nu \equiv \mathbb{E}_0\theta(\tau_1) - \theta(\mu_\tau) = \tfrac{1}{2}\theta''(\mu_\tau)\,\sigma_\tau^2 + o(\sigma_\tau^2), \qquad J(\phi) = \phi\,\nu. \tag{C.15}$$

At the calibration of Section 6.2.3 the exact value is $\nu = 0.00266$ and the approximation gives 0.00264.

Proposition C.2 (Risk neutrality)**.** *Suppose* $\gamma = 0$ *and local inaction holds. Then*

$$\phi_I^* = \frac{F_{I,0} + \kappa + \delta\mu_F}{H(\mu_\tau) - \delta\nu}, \tag{C.16}$$

any mean-preserving spread of τ_1 *raises* ν *and therefore the cutoff, and the distribution of* $F_{I,1}$ *affects the cutoff only through* μ_F*.*

Proof. With $\gamma = 0$, $P = 0$ and (C.14) is $D_0(\phi) = Z_0(\phi) - \kappa + \delta[\bar{Z}_1(\phi) - \phi\nu]$, linear in ϕ with slope $H(\mu_\tau) - \delta\nu = [\alpha - \theta(\tau_0)] + \delta[\alpha - \mathbb{E}_0\theta(\tau_1)] > 0$ by (C.2). Solving $D_0 = 0$ gives (C.16). A mean-preserving spread of τ_1 raises $\mathbb{E}_0\theta(\tau_1)$, strictly because θ is strictly convex (Rothschild and Stiglitz, 1970), hence ν and the cutoff. $F_{I,1}$ enters (C.16) only through μ_F. □

A risk-neutral firm in the inaction zone therefore responds to trade-cost uncertainty by delaying subsidiary formation, not because waiting has option value (no firm here expects to switch) but because a firm serving the destination from abroad adjusts its sales to the realised cost, so a spread in the cost raises its expected profit.

The risk premia. For a payoff Y with small variance, expanding u to second order around $\mathbb{E}_0[Y]$ on both sides of $u(CE_0(Y)) = \mathbb{E}_0[u(Y)]$ gives the Arrow–Pratt approximation

$$RP(Y) = \tfrac{1}{2} R\big(\mathbb{E}_0[Y]\big)\,\mathrm{Var}_0(Y) + o\big(\mathrm{Var}_0(Y)\big), \qquad R(\Pi) \equiv -\frac{u''(\Pi)}{u'(\Pi)} = \frac{\gamma}{\Pi}.$$

To first order a shock ε_τ changes a waiter's profit by $\phi\theta'\varepsilon_\tau$, so $\mathrm{Var}_0(\Pi_{X,1}) = (\phi\theta')^2\sigma_\tau^2 + o(\sigma_\tau^2)$; a parent's profit moves one for one with the subsidiary cost, so $\mathrm{Var}_0(\Pi_{I,1}) = \sigma_F^2$ exactly. Hence

$$RP\big(\Pi_{X,1}\big) \simeq \frac{1}{2}\,\frac{\gamma}{\bar{\Pi}_X(\phi)}\big[\phi\theta'(\mu_\tau)\big]^2\sigma_\tau^2, \qquad RP\big(\Pi_{I,1}\big) \simeq \frac{1}{2}\,\frac{\gamma}{\bar{\Pi}_I(\phi)}\,\sigma_F^2, \tag{C.17}$$

where $\bar{\Pi}_X(\phi) \equiv \pi_D(\phi) + \phi\theta(\mu_\tau)$ and $\bar{\Pi}_I(\phi) \equiv \pi_D(\phi) + \alpha\phi - \mu_F$ are total profits at expected costs (replacing expected profit by profit at expected costs changes the first premium only at higher order, since the two differ by $J = O(\sigma_\tau^2)$).

The gain for small risks. Substituting (C.15) and (C.17) into (C.14),

$$D_0(\phi) \simeq \underbrace{Z_0(\phi) - \kappa + \delta\,\bar{Z}_1(\phi)}_{\text{known costs}} + \frac{\delta}{2}\Big[\Lambda_\tau(\phi)\,\sigma_\tau^2 - \frac{\gamma}{\bar{\Pi}_I(\phi)}\,\sigma_F^2\Big], \qquad \Lambda_\tau(\phi) \equiv \underbrace{\frac{\gamma}{\bar{\Pi}_X(\phi)}\big[\phi\theta'(\mu_\tau)\big]^2}_{\text{precaution}} - \underbrace{\phi\theta''(\mu_\tau)}_{\text{convexity}}. \tag{C.18}$$

The two variances enter separately and the correlation between the shocks does not matter, because under local inaction each certainty equivalent involves one shock only. With switching, both payoffs depend on the joint distribution of $(\tau_1, F_{I,1})$ and covariances can matter. Since Z_0, $\bar{Z}_1$ and $CE_0(\Pi_{I,1})$ do

not depend on σ_τ^2, $\partial D_0/\partial\sigma_\tau^2 = \frac{\delta}{2}\Lambda_\tau + o(1)$, and the implicit-function theorem applied to $D_0(\phi_I^*) = 0$ gives

$$\frac{\partial\phi_I^*}{\partial\sigma_\tau^2} = -\frac{\delta}{2}\frac{\Lambda_\tau(\phi_I^*)}{D_{0,\phi}} + o(1), \tag{C.19}$$

where $D_{0,\phi} > 0$ is the slope of the gain in capability at the marginal firm, continuous in σ_τ^2 under local inaction and equal to $H(\mu_\tau)$ at $\sigma_\tau^2 = 0$ (Appendix C.1). Let

$$s(\phi;\tau) \equiv \frac{\phi\theta(\tau)}{\pi_D(\phi) + \phi\theta(\tau)} \in (0,1] \tag{C.20}$$

be the destination's share of the firm's total profit at trade cost τ; $s(\phi;\mu_\tau)$ is the share in (11).

Proposition C.3 (Precautionary subsidiary formation)**.** *Under local inaction, with $\sigma_F = 0$:*

(i) for small trade-cost risk,

$$\left.\frac{\partial\phi_I^*}{\partial\sigma_\tau^2}\right|_{\sigma_\tau^2=0} = -\frac{\delta\,\Lambda_\tau(\phi_I^*)}{2H(\mu_\tau)}, \qquad \textit{which is negative if and only if} \qquad \gamma\, s(\phi_I^*;\mu_\tau) > \frac{\sigma}{\sigma-1}; \tag{C.21}$$

(ii) if $\gamma\, s(\phi_I^*;\tau) \geq \sigma/(\sigma-1)$ for all τ in the support of τ_1, any mean-preserving spread of τ_1 weakly lowers the cutoff; when $\pi_D > 0$ it suffices to check the upper end of the support.

Proof. (i) Substituting the CES derivatives into Λ_τ,

$$\Lambda_\tau(\phi) = \frac{\phi\theta(\mu_\tau)(\sigma-1)}{\mu_\tau^2}\Big[(\sigma-1)\gamma\, s(\phi;\mu_\tau) - \sigma\Big],$$

and the factor outside the brackets is positive, so $\Lambda_\tau > 0 \iff \gamma s > \sigma/(\sigma-1)$. The share appears because the risk premium divides by total profit while the convexity term does not. Evaluating (C.19) at $\sigma_\tau^2 = 0$, where $D_{0,\phi} = H(\mu_\tau)$, gives (C.21); at equality higher-order terms determine the sign. With thin-tailed shocks the requirement $|\tilde{Z}_1| < \kappa$ in all states can be relaxed to $|\tilde{Z}_1(\phi_I^*)| < \kappa$, since switching probabilities are then exponentially small in $1/\sigma_\tau^2$ and do not affect the second-order result.

(ii) Let $v(\tau) \equiv u(X(\tau))$ with $X(\tau) \equiv \pi_D(\phi) + \phi\theta(\tau)$. Then

$$v''(\tau) = u'(X(\tau))\{\phi\theta''(\tau) - R(X(\tau))[\phi\theta'(\tau)]^2\},$$

so by the same algebra with μ_τ replaced by τ, v is concave at τ if and only if $\gamma s(\phi;\tau) \geq \sigma/(\sigma-1)$. If this holds on an interval containing the support of the riskier distribution, a mean-preserving spread weakly lowers $\mathbb{E}_0[v(\tau_1)]$ (Rothschild and Stiglitz, 1970), so $CE_0(\Pi_{X,1})$ weakly falls while $CE_0(\Pi_{I,1})$ is unchanged; D_0 weakly rises at the old cutoff and, since D_0 is increasing in ϕ, the cutoff weakly falls. When $\pi_D > 0$, $s(\phi;\tau)$ falls with τ, so the upper end of the support is the binding check. If the reverse strict inequality holds throughout the support the ranking is reversed. □

Proposition C.2 is the special case $\gamma = 0$: the condition in (C.21) then fails for every share, so uncertainty raises the cutoff. Part (i) is (11) in the text.

Exposure and capability. If background activities carry a fixed overhead, $\pi_D(\phi) = B_D\phi - f_D$ with $f_D > 0$, then

$$s(\phi;\mu_\tau) = \frac{\theta(\mu_\tau)\,\phi}{[B_D + \theta(\mu_\tau)]\phi - f_D}, \qquad \frac{ds}{d\phi} = -\frac{\theta(\mu_\tau) f_D}{\left\{[B_D + \theta(\mu_\tau)]\phi - f_D\right\}^2} < 0, \tag{C.22}$$

so less capable firms are more exposed to a single destination and need less risk aversion for precaution. This is the first of the size channels mentioned in Section 6.2.2; the second is (13) below.

Corollary C.1 (Subsidiary-cost uncertainty)**.** *Under local inaction, with $\sigma_\tau = 0$, any mean-preserving spread in $F_{I,1}$ weakly raises the cutoff, strictly if $\gamma > 0$. For small risk,*

$$\left.\frac{\partial \phi_I^*}{\partial \sigma_F^2}\right|_{\sigma_F^2=0} = \frac{\delta\,\gamma}{2\,\bar{\Pi}_I(\phi_I^*)\,H(\mu_\tau)} > 0. \tag{C.23}$$

Proof. With $\sigma_\tau = 0$, $\Pi_{X,1}$ is deterministic and $u(\pi_D(\phi) + \alpha\phi - F_I)$ is strictly concave in F_I for $\gamma > 0$, so a mean-preserving spread in $F_{I,1}$ lowers $CE_0(\Pi_{I,1})$ and raises the cutoff; for $\gamma = 0$ it has no effect. For small risk, (C.18) gives $\partial D_0/\partial\sigma_F^2 = -\delta\gamma/(2\bar{\Pi}_I)$, and the implicit-function theorem with $D_{0,\phi} = H(\mu_\tau)$ at $\sigma_F^2 = 0$ gives (C.23). □

Fixed and variable frictions: derivation of (13). As in the notation of Appendix C.1, let a share of the friction of serving from abroad be a per-period fixed cost driven by the same innovation as the trade cost, $F_{X,1} = \mu_X + k\varepsilon_\tau$ with $k \geq 0$, so that a waiter's profit is $\Pi_{X,1} = \pi_D + \phi\theta(\mu_\tau + \varepsilon_\tau) - \mu_X - k\varepsilon_\tau$. To first order a shock changes it by $(\phi\theta' - k)\varepsilon_\tau$, and since $\theta' < 0$ the two components move profit in the same direction, so $\mathrm{Var}_0(\Pi_{X,1}) = (\phi|\theta'| + k)^2\sigma_\tau^2 + o(\sigma_\tau^2)$. With

$$\lambda(\phi) \equiv \frac{k}{\phi|\theta'(\mu_\tau)| + k}, \tag{C.24}$$

the fixed cost's share of the standard deviation of profit, the variance is $(\phi\theta')^2\sigma_\tau^2/(1-\lambda)^2$. The convexity term is unchanged, $\frac{1}{2}\phi\theta''\sigma_\tau^2$, because $F_{X,1}$ enters profit linearly. Hence $\Lambda_\tau = (\gamma/\bar{\Pi}_X)(\phi\theta')^2/(1-\lambda)^2 - \phi\theta''$ with $\bar{\Pi}_X$ now net of μ_X, and the factorisation in the proof of Proposition C.3 gives $\Lambda_\tau > 0 \iff \gamma s > (1-\lambda)^2\sigma/(\sigma-1)$, which is (13). Since $\lambda(\phi)$ is decreasing in ϕ for given k, the threshold is lower for less capable firms. If the two shocks are imperfectly correlated the variance has a cross term and the same condition holds with $(1-\lambda)^{-2}$ replaced by $\mathrm{Var}_0(\Pi_{X,1})/[(\phi\theta')^2\sigma_\tau^2]$.

Expected costs under uncertainty. The comparative statics of Proposition C.1 with respect to expected costs survive uncertainty and risk aversion.

Proposition C.4 (Expected costs under uncertainty)**.** *Suppose local inaction holds before and after the change and D_0 is increasing in ϕ. Then, for any $\gamma \geq 0$ and any distribution of the shocks, ϕ_I^* is non-increasing in μ_τ and non-decreasing in μ_F.*

Proof. Under local inaction a higher μ_τ lowers $\Pi_{X,1} = \pi_D + \phi\theta(\mu_\tau + \varepsilon_\tau)$ in every state and leaves $\Pi_{I,1}$ unchanged. The new payoff is first-order stochastically dominated by the old one, and since u is increasing

$CE_0(\Pi_{X,1})$ falls. Hence D_0 rises at every ϕ and, since D_0 is increasing in ϕ, the cutoff weakly falls. A higher μ_F lowers $\Pi_{I,1}$ in every state and raises the cutoff. □

C.5 Outside the inaction zone: the option term (Section 6.2.3)

Position of the marginal firm. With M as in (C.10) and $\hat{m} \equiv \mathbb{E}_0[\tilde{Z}_1] = \bar{Z}_1 - J$, the cutoff condition $Z_0 - \kappa + \delta(\bar{Z}_1 - J + P - W) = 0$ gives

$$\bar{Z}_1(\phi_I^*) = \frac{\kappa + M + \delta(J - P + W)}{1+\delta}, \qquad \hat{m}(\phi_I^*) = \frac{\kappa + M - J + \delta(W - P)}{1+\delta}. \tag{C.25}$$

Because κ is paid up front, $\hat{m} > 0$ unless local presence is expected to lose more than κ of its current advantage or the precautionary term is large: the marginal firm typically expects local presence to be profitable at $t = 1$.

Lemma C.3 (The option term)**.** *Suppose the realised advantage can be written as $\tilde{Z}_1 = \hat{m} + \ell\xi$, where $\hat{m} = \mathbb{E}_0[\tilde{Z}_1]$, $\ell > 0$ is its standard deviation and ξ is a standardised shock with mean zero, unit variance and a distribution symmetric around zero. If $\hat{m} > 0$, then $W \geq 0$ and $dW/d\ell \geq 0$: greater dispersion raises the value of forming a subsidiary later by more than the value of reversing an existing one.*

Proof. The option values are $\Omega_X(\ell) = \mathbb{E}_0[(\hat{m} + \ell\xi - \kappa)^+]$ and $\Omega_I(\ell) = \mathbb{E}_0[(-\hat{m} - \ell\xi - \kappa)^+]$. Let $\mathcal{J}(c) \equiv \mathbb{E}_0[\xi\,\mathbf{1}\{\xi > c\}]$; by symmetry $\mathcal{J}(-c) = \mathcal{J}(c)$, and $\mathcal{J}$ is weakly decreasing on $[0,\infty)$. Differentiating, and using symmetry for Ω_I,

$$\frac{d\Omega_X}{d\ell} = \mathcal{J}\Big(\frac{\kappa - \hat{m}}{\ell}\Big), \qquad \frac{d\Omega_I}{d\ell} = \mathcal{J}\Big(\frac{\kappa + \hat{m}}{\ell}\Big).$$

If $\hat{m} > 0$ then $\kappa + \hat{m} \geq |\kappa - \hat{m}|$, so $dW/d\ell = d\Omega_X/d\ell - d\Omega_I/d\ell \geq 0$, strictly if ξ has mass between $|\kappa - \hat{m}|/\ell$ and $(\kappa + \hat{m})/\ell$. The same comparison gives $W \geq 0$. □

For trade-cost risk the representation holds to first order, with $\ell = \phi|\theta'(\mu_\tau)|\sigma_\tau$; for subsidiary-cost risk it is exact, with $\ell = \sigma_F$. The lemma concerns the option term only: the precautionary term may also change with ℓ, and under a spread in τ_1 so does the convexity term. Under the conditions of the lemma, greater dispersion raises the value of waiting and works against immediate subsidiary formation; whether it outweighs the precautionary term is the question Figure 8 answers numerically.

When the option term starts to matter. To first order the standard deviation of $\tilde{Z}_1$ under trade-cost risk is $\phi|\theta'(\mu_\tau)|\sigma_\tau$, and the distance of the marginal firm from the upper edge of the inaction zone is $\kappa - \mathbb{E}_0\tilde{Z}_1$. The option term becomes noticeable once the two are comparable, at $\sigma_\tau \approx (\kappa - \mathbb{E}_0\tilde{Z}_1)/(\phi_I^*|\theta'(\mu_\tau)|)$: the region in which precaution operates alone widens with κ and shrinks as an expected deterioration in exporting conditions moves the marginal firm towards the edge. In the benchmark of Figure 8, where half of the friction is fixed, the distance to the edge is $\kappa - \mathbb{E}_0\tilde{Z}_1 = 0.119$ with the baseline switching cost of two years, where the option term becomes visible from a dispersion of about 0.04 (the bad tail of the shock reaches the edge before its mean does) and slows the rise of subsidiary formation without reversing it; 0.190 with the upper bound of 2.9 years, where the option term stays nil over the whole range; 0.081 with 1.6 years, where the response peaks at $\sigma_\tau = 0.036$ and returns to zero at 0.061; and 0.032 with one year,

where it turns negative at 0.015. With a switching cost of half a year the marginal firm is outside the zone already under expected costs, and uncertainty lowers subsidiary formation at every dispersion.

C.6 Calibration, numerical method and experiments (Section 6.2.3)

Parameters. The model is solved in period units: current profit enters with weight one and period-1 profit with weight $\delta = 1$. Shocks are normal, expectations are computed by Gauss–Hermite quadrature (30 nodes for τ_1, 20 for $F_{I,1}$; 80 nodes for τ_1 in Figure 8, where the tail of the shock crosses the switching threshold), certainty equivalents are computed from (C.5) on the quadrature nodes, and cutoffs solve $D_0(\phi) = 0$ by bracketing.

Normalisations. Before the referendum there is no expected change in costs, $\mu_\tau = \tau_0$ and $\mu_F = F_{I,0}$, and no uncertainty. Capability is normalised so that the marginal firm in this situation has $\phi = 1$. Its export profit per period at status-quo costs is $\theta(\tau_0) = 0.75$, and $\kappa = 0.45 \times 0.75 = 0.34$; since period 0 is the 4.5-year transition, this is about two years of that firm's annual export profit. $F_{I,0}$ makes this firm exactly indifferent: by (C.11) with $M = 0$, $(1+\delta)([\alpha - \theta(\tau_0)] - F_{I,0}) = \kappa$, so $F_{I,0} = [\alpha - \theta(\tau_0)] - \kappa/(1+\delta) = 0.080$, about 8% of the subsidiary's variable profit. With a Pareto distribution of shape k and the pre-referendum cutoff at one, the number of new parents relative to its pre-referendum level is $(\phi_I^*)^{-k}$; the shape rescales every entry of Table 5 and the vertical axes of Figures 8 and 9 without changing any sign or ranking, since the cutoffs do not depend on it.

The inaction zone in the experiments. Under expected Brexit costs with the baseline switching cost and a proportional friction, (C.11) gives $\phi^z = 0.816$ and $M(\phi^z) = 0.092 < \delta\kappa = 0.338$, so the marginal firm is in the zone with $\bar{Z}_1 = 0.215$; because ϕ^z falls as μ_τ rises, $M(\phi^z)$ stays below $\delta\kappa$ for any μ_τ. Since $F_{I,0}$ is backed out from κ, $F_{I,0} + \kappa/(1+\delta)$ and hence ϕ^z are the same for every switching cost; the cases of Figure 8 differ in $\delta\kappa$, 0.481, 0.263 and 0.165 for 2.9, 1.6 and one year, all above $M(\phi^z) = 0.092$, so the marginal firm is in the zone under expected costs in all four cases. With half of the friction fixed the known-cost cutoff is 0.797 and the same holds, with the distances to the edge given in Appendix C.5. With a switching cost of half a year, $\delta\kappa = 0.083$ and the third case of (C.12) applies for expected rises in trade costs above 4.8%; under expected Brexit costs the cutoff is then $\phi^+ = 0.834$ and no longer responds to μ_τ.

Admissible realisations. A normal shock is unbounded, while u in (C.5) requires positive profit and θ a positive trade cost. In the quadrature the shock takes only the node values, the most extreme of which lie 9.7 standard deviations from the mean with 30 nodes and 16.8 with 80, with weights of 10^{-21} and 10^{-62}. At every cutoff reported in Table 5, period-1 profit is positive at every node in both modes (the smallest value is 0.70, for a UK waiter with half of the friction fixed) and the trade cost stays positive (its smallest node value is 0.87, with the proportional friction). The restrictions $\tau \geq 1$ and $\alpha > \theta(\tau)$ of Section 6.1 are therefore to be read on the admissible support: they fail only at the most extreme quadrature node, 9.7 standard deviations below the mean (where $\tau = 0.87$ with the proportional friction and 0.997 with half of it fixed), of weight 10^{-21}, and hold at every node within four standard deviations of the mean, where $\tau \geq 1.02$ and $\theta(\tau) \leq 0.94 < \alpha$ in every experiment of Section 6.2.3, including the largest dispersion of Figure 8. The

root-finding routine evaluates D_0 only at capabilities for which profit after any switch is positive at every node, raising the lower end of the bracket otherwise, so that utility is never evaluated on an inadmissible realisation. Truncating both shocks at four standard deviations, on which support the assumptions hold in every state, and recomputing expectations with Gauss–Legendre quadrature on the truncated support leaves every entry of Table 5 unchanged to two decimals; truncating at three standard deviations changes the response to uncertainty by at most 0.01 and sets the option term to zero in the first row, since the realisations that would make a firm switch lie beyond that range. The lines of Figure 8 move by at most 0.05 under either truncation.

Experiments. In Table 5 the $\bar{Z}_1$ column is the cutoff with known costs at the expected post-referendum costs; the next three add trade-cost uncertainty with $\gamma = 0$ and no switching, set $\gamma = 4$ with no switching, and allow switching; each cutoff is converted into a change in the number of new parents with the Pareto shape defined in section 6, and the columns report differences. The table is therefore a sequential numerical accounting: because the cutoff and the Pareto transformation are nonlinear, the entries in the J, P and W columns are not the terms of (9) evaluated at a fixed capability, and their sum equals the total response by construction. In the rows with a fixed friction, let $L \equiv \theta(\tau_0) - \theta(\mu_\tau) = 0.112$ be the expected Brexit loss of export profit of the pre-referendum marginal firm and $\omega \equiv |\theta'(\mu_\tau)|\sigma_\tau = 0.050$ its standard deviation. The fixed cost of serving from abroad is zero at $t = 0$ and $F_{X,1} = \lambda L + \lambda\omega\xi$ at $t = 1$, with ξ the standardised trade-cost shock, so that $\mu_X = \lambda L = 0.056$ and $k\varepsilon_\tau = \lambda\omega\xi$; the expected trade cost is lowered to μ'_τ with $\theta(\mu'_\tau) = \theta(\tau_0) - (1-\lambda)L$ ($\mu'_\tau = 1.129$), and the trade-cost shock is scaled so that $|\theta'(\mu'_\tau)|\sigma'_\tau = (1-\lambda)\omega$ ($\sigma'_\tau = 0.0136$). For the pre-referendum marginal firm the expected loss and the profit risk are therefore the same as in the proportional case, with a share $\lambda = \frac{1}{2}$ of each moved to the fixed cost. Because the post-referendum marginal firm is smaller than the pre-referendum one, the fixed share of its own profit risk is higher, $\lambda(\phi_I^*) = 0.56$ by (C.24); this is the value at which (13) is evaluated for the thresholds in Section 6 and for the dotted line in Figure 9, where it coincides with the zero contour of the numerical solution ($\gamma s = 0.26$ with half of the friction fixed, 1.33 with a proportional friction). In Panel B of Table 5 the second row raises μ_F by 0.136, the value at which subsidiary formation with known costs is 4.4% below its pre-referendum level, and the third adds $\sigma_F = 0.02$. Figure 9 recomputes the last column of Table 5 on a grid of (γ, s) with the baseline switching cost, $\sigma_F = 0$ and $\mu_F = F_{I,0}$.